• PROJECT SCHEMATIC / 3D HOLOGRAPHIC BLUEPRINTS

JARVIS EPISODE 31 GUIDE

by Pulse of AI

▲ CRITICAL WARNING: MANDATORY PRE-REQUISITE

You must watch all the previous videos (Parts 1 to 30) and extract the baseline code and environment files directly from their pinned comments. If you do not configure the previous dependencies, JARVIS might not work!

01. COMPLETE SERIES ROADMAP (WATCH IN CHRONOLOGICAL ORDER)

MODULE	ENGINEERING CAPABILITY ADDED	DIRECT SOURCE
PART 1	Core Brain Setup & Instant Live API Integration	■ Watch Ep 1
PART 2	Autonomous System Expansion & Pipeline Upgrades	■ Watch Ep 2
PART 3	Autonomous System Expansion & Pipeline Upgrades	■ Watch Ep 3
PART 4	Autonomous System Expansion & Pipeline Upgrades	■ Watch Ep 4
PART 5	Autonomous System Expansion & Pipeline Upgrades	■ Watch Ep 5
PART 6	Autonomous System Expansion & Pipeline Upgrades	■ Watch Ep 6
PART 7	Autonomous Desktop Tool Calling Active	■ Watch Ep 7
PART 8	Autonomous System Expansion & Pipeline Upgrades	■ Watch Ep 8
PART 9	Autonomous System Expansion & Pipeline Upgrades	■ Watch Ep 9
PART 10	Autonomous System Expansion & Pipeline Upgrades	■ Watch Ep 10
PART 11	Autonomous System Expansion & Pipeline Upgrades	■ Watch Ep 11
PART 12	Autonomous System Expansion & Pipeline Upgrades	■ Watch Ep 12
PART 13	Autonomous System Expansion & Pipeline Upgrades	■ Watch Ep 13
PART 14	Autonomous System Expansion & Pipeline Upgrades	■ Watch Ep 14
PART 15	Autonomous System Expansion & Pipeline Upgrades	■ Watch Ep 15
PART 16	Autonomous System Expansion & Pipeline Upgrades	■ Watch Ep 16
PART 17	Autonomous System Expansion & Pipeline Upgrades	■ Watch Ep 17
PART 18	Autonomous System Expansion & Pipeline Upgrades	■ Watch Ep 18
PART 19	Autonomous System Expansion & Pipeline Upgrades	■ Watch Ep 19

PART 20	Autonomous System Expansion & Pipeline Upgrades	■ Watch Ep 20
PART 21	Autonomous System Expansion & Pipeline Upgrades	■ Watch Ep 21
PART 22	Autonomous System Expansion & Pipeline Upgrades	■ Watch Ep 22
PART 23	Autonomous Serious Mode Pipeline	■ Watch Ep 23
PART 24	Autonomous System Expansion & Pipeline Upgrades	■ Watch Ep 24
PART 25	Autonomous System Expansion & Pipeline Upgrades	■ Watch Ep 25
PART 26	Autonomous System Expansion & Pipeline Upgrades	■ Watch Ep 26
PART 27	Autonomous System Expansion & Pipeline Upgrades	■ Watch Ep 27
PART 28	Autonomous System Expansion & Pipeline Upgrades	■ Watch Ep 28
PART 29	Project Doppelgänger (Ghost Operator)	■ Watch Ep 29
PART 30	Project Sentinel (3D Holographic Globe HUD)	■ Watch Ep 30
PART 31	**Project Schematic (3D Holographic Blueprints)**	■ Watch Ep 31

02. ENVIRONMENT ARCHITECTURE & SYSTEM CONFIG

[1] Terminal Dependencies:
pip install google-genai pyaudio geopy plotly PyQt6 PyQt6-WebEngine flask pywebview requests

[2] Secure Environment File (.env):
GOOGLE_API_KEY=your_gemini_api_key

[3] Episode 31 Core Innovation:
Integration of a Three.js web engine powered by a Flask backend and pywebview to render a borderless, 100% transparent interactive 3D holographic widget directly over your Windows desktop.

03. PRODUCTION CODEBASE (DYNAMICALLY EXTRACTED)

The following source code has been extracted directly from your local project files.

// MODULE: schematic_hud.py (PROJECT SCHEMATIC FLASK BACKEND & CONTROLLER)

```
# -*- coding: utf-8 -*-
"""
schematic_hud.py — JARVIS :: Project Schematic (v1 — Three.js Edition)
Floating transparent 3D holographic blueprint viewer.
Voice-activated via "JARVIS, pull up a 3D schematic of [object]".

Architecture
■■■■■■■■■■■■■
 Python ■■[Flask:9393]■■■ HTML/Three.js in pywebview transparent subprocess
 agent.py calls launch_schematic_hud(query="airplane engine") as a Gemini tool.

Built for Swayam Dhawale — Pulse of AI.
"""

import os
import sys
import json
import time
import threading
import subprocess
import logging
from typing import Optional

# ■■ Optional imports ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
try:
    import requests as _requests
    _HAS_REQUESTS = True
except ImportError:
    _HAS_REQUESTS = False

try:
    import webview as _webview
    _HAS_WEBVIEW = True
except ImportError:
    _HAS_WEBVIEW = False

try:
    from flask import Flask, jsonify, render_template_string, request as _flask_req
    _HAS_FLASK = True
except ImportError:
    _HAS_FLASK = False

# ■■ Constants ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
FLASK_PORT = 9393 # Unique port — no conflict with orbital (9191) or globe (9292)
_flask_thread: Optional[threading.Thread] = None
_server_started = False
_current_schematic: dict = {}
_viewer_process: Optional[subprocess.Popen] = None

# ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
# SEMANTIC QUERY RESOLVER
# Maps natural language queries to blueprint configurations.
# ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■

_SCHEMATICS: dict = {
    # ■■ Aerospace ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
    "airplane engine": {
        "id": "turbofan",
        "label": "CFM LEAP-1B Turbofan Engine",
        "type": "aerospace",
        "color_primary": "#00e5ff",
```

```
        "color_secondary": "#0051ff",
        "color_accent": "#ff6d00",
        "specs": [
            "Bypass Ratio: 11:1",
            "Thrust: 120 kN",
            "Core Temp: 1,650 °C",
            "Fan Diameter: 1.76 m",
            "N1 Speed: 3,800 RPM",
            "Compression Ratio: 40:1",
        ],
        "description": "High-bypass turbofan with composite fan blades, advanced combustor, and full-authority
digital engine control.",
        "voice_intro": "Rendering CFM LEAP-1B turbofan engine schematic, sir. Displaying high-pressure turbine
stages and bypass duct architecture.",
    },
    "turbofan": {
        "id": "turbofan",
        "label": "CFM LEAP-1B Turbofan Engine",
        "type": "aerospace",
        "color_primary": "#00e5ff",
        "color_secondary": "#0051ff",
        "color_accent": "#ff6d00",
        "specs": ["Bypass Ratio: 11:1", "Thrust: 120 kN", "Core Temp: 1,650 °C", "Fan Diameter: 1.76 m"],
        "description": "High-bypass turbofan engine used in Boeing 737 MAX and Airbus A320neo aircraft.",
        "voice_intro": "Turbofan engine blueprint online, sir.",
    },
    "rocket": {
        "id": "rocket",
        "label": "Multi-Stage Orbital Launch Vehicle",
        "type": "aerospace",
        "color_primary": "#00e5ff",
        "color_secondary": "#ffffff",
        "color_accent": "#ff1744",
        "specs": [
            "Stage 1 Engines: 9 × Merlin 1D",
            "Thrust (Sea Level): 7,607 kN",
            "Payload to LEO: 22,800 kg",
            "Height: 70 m",
            "Diameter: 3.66 m",
            "Fuel: RP-1 / Liquid Oxygen",
        ],
        "description": "Two-stage orbital rocket with reusable first stage booster and deployable payload fairing.",
        "voice_intro": "Multi-stage orbital launch vehicle schematic initialized, sir. Displaying stage separation
systems and engine cluster.",
    },
    "drone": {
        "id": "drone",
        "label": "Advanced Reconnaissance Quadcopter",
        "type": "aerospace",
        "color_primary": "#00e5ff",
        "color_secondary": "#76ff03",
        "color_accent": "#ffd600",
        "specs": [
            "Motors: 4× Brushless 2306 2400KV",
            "Max Speed: 140 km/h",
            "Flight Time: 35 min",
            "Max Range: 12 km",
            "Camera: 4K / 60fps Gimbal",
            "Payload Capacity: 500 g",
        ],
        "description": "X-frame carbon fiber reconnaissance quadcopter with active thermal imaging and autonomous
navigation.",
        "voice_intro": "Quadcopter reconnaissance drone schematic online, sir.",
    },
    # ■■ Energy & Power ■■■
```

```
        "type": "energy",
        "color_primary": "#00e5ff",
        "color_secondary": "#e040fb",
        "color_accent": "#ffffff",
        "specs": [
            "Output: 3 GJ/sec",
            "Core Element: Palladium (Mk.II)",
            "Coil Count: 12 Electromagnetic",
            "Plasma Temp: 8,100 K",
            "Diameter: 12.7 cm",
            "Power Density: 8.1 TW/kg",
        ],
        "description": "Miniaturized arc reactor based on Howard Stark's original toroidal field generator.
Palladium-core with 12 copper electromagnetic induction coils.",
        "voice_intro": "Mark II Arc Reactor schematic deployed, sir. Palladium core nominal. Electromagnetic coil
array ready for inspection.",
    },
    "arc reactor mark ii": {
        "id": "arc_reactor",
        "label": "Mark II Arc Reactor — Palladium Core",
        "type": "energy",
        "color_primary": "#00e5ff",
        "color_secondary": "#e040fb",
        "color_accent": "#ffffff",
        "specs": ["Output: 3 GJ/sec", "Core Element: Palladium (Mk.II)", "Coil Count: 12"],
        "description": "Miniaturized arc reactor — Mark II edition.",
        "voice_intro": "Mark II Arc Reactor schematic online, sir.",
    },
    # ■■ Automotive ■■■
```

```
        "specs": [
            "Qubits: 72 (Sycamore architecture)",
            "Gate Fidelity: 99.87%",
            "Operating Temp: 15 mK",
            "Coherence Time: 300 µs",
            "Gate Speed: 12 ns (single-qubit)",
            "Connectivity: Heavy-hex lattice",
        ],
        "description": "Superconducting transmon qubit processor operating at millikelvin temperatures inside a dilution refrigerator. Gold wire-bonded to cryogenic control electronics.",
        "voice_intro": "Seventy-two qubit quantum processor schematic initialized, sir. Operating at fifteen millikelvin. Displaying cryo-coil architecture and qubit lattice.",
    },
    "processor": {
        "id": "quantum_cpu",
        "label": "72-Qubit Superconducting Quantum Processor",
        "type": "computing",
        "color_primary": "#00e5ff",
        "color_secondary": "#7c4dff",
        "color_accent": "#64ffda",
        "specs": ["Qubits: 72", "Operating Temp: 15 mK", "Gate Fidelity: 99.87%"],
        "description": "Superconducting quantum processor.",
        "voice_intro": "Quantum processor schematic online, sir.",
    },
    # ■■ Biology & Medical ■■■
```

```
            "Communication: Encrypted tri-band",
            "Sensor Suite: Thermal / EM / LiDAR",
            "Impact Rating: Mach 2.5 ballistic",
        ],
        "description": "Mark LXXXV helmet with retractable faceplate nanocomposite, integrated F.R.I.D.A.Y. AI visor, and autonomous threat-assessment sensor suite.",
        "voice_intro": "Iron Man Mark Eighty-Five helmet schematic online, sir. Faceplate assembly and visor array ready for inspection.",
    },
    "satellite": {
        "id": "satellite",
        "label": "Low Earth Orbit Communication Satellite",
        "type": "aerospace",
        "color_primary": "#00e5ff",
        "color_secondary": "#b0bec5",
        "color_accent": "#ffd600",
        "specs": [
            "Orbit Altitude: 550 km LEO",
            "Solar Panels: 4 deployable (8 kW)",
            "Comms: Ku-band phased array",
            "Thruster: Hall-effect ion propulsion",
            "Mass: 260 kg",
            "Lifespan: 7 years",
        ],
        "description": "Flat-panel LEO communication satellite with four deployable solar arrays, electronically steered antenna, and redundant attitude-control thrusters.",
        "voice_intro": "Low Earth Orbit communication satellite schematic online, sir. Displaying solar panel deployment and phased array antenna.",
    },
}

# Aliases & partial match lookup
_ALIASES = {
    "plane engine": "airplane engine",
    "jet engine": "airplane engine",
    "turbine": "airplane engine",
    "jet": "airplane engine",
    "aircraft engine": "airplane engine",
    "arc": "arc reactor",
    "reactor": "arc reactor",
    "iron man reactor": "arc reactor",
    "quadcopter": "drone",
    "quad": "drone",
    "uav": "drone",
    "v8": "v8 engine",
    "car engine": "v8 engine",
    "quantum": "quantum cpu",
    "cpu": "quantum cpu",
    "chip": "quantum cpu",
    "quantum computer": "quantum cpu",
    "cybernetic": "cybernetic heart",
    "iron man": "iron man helmet",
    "mark 85": "iron man helmet",
    "helmet": "iron man helmet",
    "suit": "iron man helmet",
    "space satellite": "satellite",
    "comsat": "satellite",
}


def resolve_schematic(query: str) -> dict:
    """
    Maps a natural language query to a blueprint configuration dict.
    Falls back to 'turbofan' (airplane engine) if nothing matches.
    """
    q = query.lower().strip()

    # Direct match
```

```
    if q in _SCHEMATICS:
        return _SCHEMATICS[q]

    # Alias lookup
    if q in _ALIASES:
        return _SCHEMATICS[_ALIASES[q]]

    # Partial match — check if query contains any schematic key
    for key in _SCHEMATICS:
        if key in q or q in key:
            return _SCHEMATICS[key]

    # Partial alias match
    for alias, target in _ALIASES.items():
        if alias in q:
            return _SCHEMATICS[target]

    # Token overlap match
    q_tokens = set(q.split())
    best_match = None
    best_score = 0
    for key in _SCHEMATICS:
        k_tokens = set(key.split())
        overlap = len(q_tokens & k_tokens)
        if overlap > best_score:
            best_score = overlap
            best_match = key
    if best_match and best_score > 0:
        return _SCHEMATICS[best_match]

    # Universal 3D CAD Generator (Gemini AI + Procedural Fallback)
    try:
        from schematic_generator import resolve_any_schematic
        return resolve_any_schematic(query)
    except Exception as exc:
        logging.error(f"Generative schematic error: {exc}")
        return _SCHEMATICS["airplane engine"]


# ═══════════════════════════════════════════════════════════════════════════
# THE THREE.JS HOLOGRAPHIC HUD TEMPLATE
# ═══════════════════════════════════════════════════════════════════════════

_HTML = r"""<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>JARVIS :: PROJECT SCHEMATIC</title>
<style>
  * { margin: 0; padding: 0; box-sizing: border-box; }

  body {
    background: #000608 !important;
    overflow: hidden;
    font-family: 'Courier New', monospace;
    -webkit-app-region: no-drag;
    user-select: none;
  }

  #canvas-container {
    position: fixed;
    inset: 0;
    background: transparent;
  }

  canvas { display: block; background: transparent !important; }
```

```
/* ■■ Drag Handle (top strip) ■■ */
#drag-bar {
  position: fixed;
  top: 0; left: 0; right: 0;
  height: 36px;
  -webkit-app-region: drag;
  background: linear-gradient(
    90deg,
    rgba(0,229,255,0.06) 0%,
    rgba(0,81,255,0.10) 50%,
    rgba(0,229,255,0.06) 100%
  );
  border-bottom: 1px solid rgba(0,229,255,0.25);
  display: flex;
  align-items: center;
  padding: 0 16px;
  gap: 12px;
  z-index: 9999;
}

.drag-dots {
  display: flex; gap: 5px;
}
.drag-dot {
  width: 8px; height: 8px; border-radius: 50%;
  background: rgba(0,229,255,0.5);
  box-shadow: 0 0 6px rgba(0,229,255,0.8);
}

#title-bar-label {
  flex: 1;
  font-size: 10px;
  letter-spacing: 3px;
  color: rgba(0,229,255,0.8);
  text-transform: uppercase;
}

/* ■■ Close Button ■■ */
#btn-close {
  -webkit-app-region: no-drag;
  width: 20px; height: 20px;
  border: 1px solid rgba(255,50,50,0.5);
  background: rgba(255,50,50,0.08);
  color: rgba(255,80,80,0.9);
  font-size: 12px;
  cursor: pointer;
  display: flex; align-items: center; justify-content: center;
  border-radius: 3px;
  transition: all 0.2s;
}
#btn-close:hover { background: rgba(255,50,50,0.3); color: #fff; }

/* ■■ Control Buttons ■■ */
#controls {
  position: fixed;
  bottom: 20px; left: 50%;
  transform: translateX(-50%);
  display: flex;
  gap: 10px;
  z-index: 9999;
  -webkit-app-region: no-drag;
}

.ctrl-btn {
  padding: 6px 16px;
  font-family: 'Courier New', monospace;
  font-size: 10px;
  letter-spacing: 2px;
```

```
  border: 1px solid rgba(0,229,255,0.4);
  background: rgba(0,20,40,0.55);
  color: rgba(0,229,255,0.85);
  cursor: pointer;
  border-radius: 3px;
  transition: all 0.2s;
  text-transform: uppercase;
  backdrop-filter: blur(4px);
}
.ctrl-btn:hover {
  background: rgba(0,229,255,0.12);
  border-color: rgba(0,229,255,0.9);
  color: #fff;
  box-shadow: 0 0 12px rgba(0,229,255,0.4);
}
.ctrl-btn.active {
  background: rgba(0,229,255,0.2);
  color: #fff;
  box-shadow: 0 0 16px rgba(0,229,255,0.6);
}

/* ■■ Tech Specs Panel (Right HUD) ■■ */
#specs-panel {
  position: fixed;
  right: 20px;
  top: 50%;
  transform: translateY(-50%);
  width: 220px;
  background: rgba(0, 8, 20, 0.55);
  border: 1px solid rgba(0,229,255,0.22);
  border-left: 3px solid rgba(0,229,255,0.6);
  padding: 14px 12px;
  z-index: 9998;
  backdrop-filter: blur(6px);
  -webkit-app-region: no-drag;
}

.spec-title {
  font-size: 9px;
  letter-spacing: 3px;
  color: rgba(0,229,255,0.5);
  text-transform: uppercase;
  margin-bottom: 10px;
  border-bottom: 1px solid rgba(0,229,255,0.15);
  padding-bottom: 6px;
}

.spec-label { font-size: 10px; color: rgba(0,229,255,0.9); margin-bottom: 6px; }
.spec-label span { color: rgba(255,255,255,0.65); }

.spec-name {
  font-size: 11px;
  color: rgba(0,229,255,1);
  font-weight: bold;
  letter-spacing: 1px;
  margin-bottom: 12px;
  line-height: 1.4;
}

/* ■■ Status Bar (bottom left) ■■ */
#status-bar {
  position: fixed;
  bottom: 56px;
  left: 20px;
  font-size: 9px;
  letter-spacing: 2px;
  color: rgba(0,229,255,0.45);
  text-transform: uppercase;
```

```
    z-index: 9998;
    -webkit-app-region: no-drag;
  }

  #status-dot {
    display: inline-block;
    width: 6px; height: 6px;
    border-radius: 50%;
    background: #00e5ff;
    box-shadow: 0 0 8px #00e5ff;
    margin-right: 6px;
    animation: pulse-dot 1.5s infinite;
  }
  @keyframes pulse-dot {
    0%,100% { opacity: 1; } 50% { opacity: 0.2; }
  }

  /* ■■ Corner brackets (sci-fi HUD corners) ■■ */
  .hud-corner {
    position: fixed;
    width: 30px; height: 30px;
    border-color: rgba(0,229,255,0.35);
    border-style: solid;
    z-index: 9000;
  }
  #c-tl { top:40px; left:0; border-width: 2px 0 0 2px; }
  #c-tr { top:40px; right:0; border-width: 2px 2px 0 0; }
  #c-bl { bottom:0; left:0; border-width: 0 0 2px 2px; }
  #c-br { bottom:0; right:0; border-width: 0 2px 2px 0; }

  /* ■■ Loading Overlay ■■ */
  #loader {
    position: fixed; inset: 0;
    background: rgba(0,5,15,0.7);
    display: flex; flex-direction: column;
    align-items: center; justify-content: center;
    z-index: 9999;
    transition: opacity 0.5s;
  }
  #loader.gone { opacity: 0; pointer-events: none; }

  .loader-ring {
    width: 60px; height: 60px;
    border: 2px solid rgba(0,229,255,0.15);
    border-top-color: #00e5ff;
    border-radius: 50%;
    animation: spin 0.8s linear infinite;
    margin-bottom: 20px;
    box-shadow: 0 0 20px rgba(0,229,255,0.4);
  }
  @keyframes spin { to { transform: rotate(360deg); } }

  .loader-text {
    font-size: 10px;
    letter-spacing: 4px;
    color: rgba(0,229,255,0.7);
    text-transform: uppercase;
    animation: flicker 2s infinite;
  }
  @keyframes flicker {
    0%,100% { opacity: 1; } 40% { opacity: 0.5; } 60% { opacity: 0.9; }
  }
</style>
</head>
<body>

<!-- HUD Frame Elements -->
<div class="hud-corner" id="c-tl"></div>
```


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```
<div class="hud-corner" id="c-tr"></div>
<div class="hud-corner" id="c-bl"></div>
<div class="hud-corner" id="c-br"></div>

<!-- Drag Bar -->
<div id="drag-bar">
  <div class="drag-dots">
    <div class="drag-dot"></div>
    <div class="drag-dot"></div>
    <div class="drag-dot"></div>
  </div>
  <div id="title-bar-label">JARVIS :: PROJECT SCHEMATIC ■ HOLOGRAPHIC BLUEPRINT SYSTEM</div>
  <button id="btn-close" onclick="window.close ? window.close() : document.body.remove()">X</button>
</div>

<!-- Three.js WebGL Canvas -->
<div id="canvas-container"></div>

<!-- Tech Specs Panel -->
<div id="specs-panel">
  <div class="spec-title">■ TECHNICAL SPECIFICATIONS</div>
  <div class="spec-name" id="spec-name">Initializing...</div>
  <div id="spec-lines"></div>
</div>

<!-- Status Bar -->
<div id="status-bar">
  <span id="status-dot"></span>
  <span id="status-text">HOLOGRAPHIC MATRIX: ONLINE</span>
</div>

<!-- Controls -->
<div id="controls">
  <button class="ctrl-btn" id="btn-explode" onclick="toggleExplode()">■ EXPLODE</button>
  <button class="ctrl-btn" id="btn-spin" onclick="toggleSpin()">■ SPIN</button>
  <button class="ctrl-btn" id="btn-xray" onclick="toggleXray()">■ X-RAY</button>
  <button class="ctrl-btn" onclick="resetCamera()">■ RESET</button>
</div>

<!-- Loader -->
<div id="loader">
  <div class="loader-ring"></div>
  <div class="loader-text">Initializing holographic matrix...</div>
</div>

<!-- Three.js r128 & OrbitControls — served locally by Flask (no internet required) -->
<script src="/three.js"></script>
<script src="/OrbitControls.js"></script>
<script>
// ■■■
```

```
const scene = new THREE.Scene();
scene.background = null;

// ■■ Camera ■■■
```

```
// ■■ Model Parts registry (for explode / xray) ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
let modelParts = []; // { mesh, originPos, explodeDir }
let modelGroup = new THREE.Group();
scene.add(modelGroup);

// ■■ Particle System ■■■■■■■■■■■■■■■■■■■
```

```
  for (let x = -size; x <= size; x += step) {
    verts.push(x, -1.2, -size, x, -1.2, size);
  }
  for (let z = -size; z <= size; z += step) {
    verts.push(-size, -1.2, z, size, -1.2, z);
  }
  geo.setAttribute('position', new THREE.BufferAttribute(new Float32Array(verts), 3));
  scene.add(new THREE.LineSegments(geo, lineMat(primaryColor, 0.08)));
})();

// ═══════════════════════════════════════════════════════════════════════════
// PROCEDURAL BLUEPRINT BUILDERS
// ═══════════════════════════════════════════════════════════════════════════

function clearModel() {
  // Remove all existing model parts
  modelParts.forEach(p => { modelGroup.remove(p.mesh); p.mesh.geometry.dispose(); });
  modelParts = [];
  // Clear any remaining children in modelGroup
  while (modelGroup.children.length > 0) {
    const c = modelGroup.children[0];
    modelGroup.remove(c);
    if (c.geometry) c.geometry.dispose();
  }
}

// ═══ TURBOFAN ENGINE ═══════════════════════════════════════════════════════
function buildTurbofan() {
  // Intake cowl
  const cowlGeo = new THREE.CylinderGeometry(0.85, 0.6, 0.4, 32, 1, true);
  const cowl = new THREE.Mesh(cowlGeo, holoMat(primaryColor, 0.18, false));
  cowl.position.set(0, 0, 0.9);
  cowl.rotation.x = Math.PI / 2;
  const cowlEdge = addEdges(cowl, primaryColor, 0.8);
  registerPart(cowlEdge, new THREE.Vector3(0, 0, 1));

  // Fan disk
  const fanGeo = new THREE.CylinderGeometry(0.8, 0.8, 0.08, 32);
  const fan = new THREE.Mesh(fanGeo, holoMat(primaryColor, 0.12));
  fan.position.set(0, 0, 1.1);
  fan.rotation.x = Math.PI / 2;
  const fanEdge = addEdges(fan, primaryColor, 0.9);
  registerPart(fanEdge, new THREE.Vector3(0, 0, 0.8));

  // Fan blades (16 blades)
  for (let i = 0; i < 16; i++) {
    const a = (i / 16) * Math.PI * 2;
    const bladeGeo = new THREE.BoxGeometry(0.07, 0.55, 0.05);
    const blade = new THREE.Mesh(bladeGeo, holoMat(primaryColor, 0.55));
    blade.position.set(
      Math.cos(a) * 0.35, Math.sin(a) * 0.35, 1.1
    );
    blade.rotation.z = a + Math.PI / 4;
    blade.rotation.x = Math.PI / 2;
    const bladeEdge = addEdges(blade, primaryColor, 0.85);
    registerPart(bladeEdge, new THREE.Vector3(Math.cos(a), Math.sin(a), 0.3));
  }

  // Core tube (bypass duct)
  const ducts = [0.88, 0.70, 0.52];
  ducts.forEach((r, i) => {
    const dGeo = new THREE.CylinderGeometry(r, r, 2.2 + i * 0.1, 32, 1, true);
    const duct = new THREE.Mesh(dGeo, holoMat(i === 0 ? primaryColor : secondaryColor, 0.10));
    duct.position.set(0, 0, 0);
    duct.rotation.x = Math.PI / 2;
    const dEdge = addEdges(duct, i === 0 ? primaryColor : secondaryColor, 0.5 - i * 0.1);
    registerPart(dEdge, new THREE.Vector3(0, 0, 0));
  });
```


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```
  // High-pressure turbine stages (3 discs)
  [-0.4, 0, 0.4].forEach((z, idx) => {
    const tGeo = new THREE.CylinderGeometry(0.42, 0.42, 0.12, 32);
    const turb = new THREE.Mesh(tGeo, holoMat(accentColor, 0.22));
    turb.position.set(0, 0, z);
    turb.rotation.x = Math.PI / 2;
    const tEdge = addEdges(turb, accentColor, 0.85);
    registerPart(tEdge, new THREE.Vector3(0, idx * 0.2, -idx * 0.3));
  });

  // Central shaft
  const shaftGeo = new THREE.CylinderGeometry(0.10, 0.10, 2.6, 16);
  const shaft = new THREE.Mesh(shaftGeo, holoMat(secondaryColor, 0.45));
  shaft.rotation.x = Math.PI / 2;
  const shaftEdge = addEdges(shaft, secondaryColor, 0.9);
  registerPart(shaftEdge, new THREE.Vector3(0, 0, 0));

  // Exhaust cone
  const exGeo = new THREE.ConeGeometry(0.5, 0.6, 32);
  const ex = new THREE.Mesh(exGeo, holoMat(accentColor, 0.22));
  ex.position.set(0, 0, -1.4);
  ex.rotation.x = -Math.PI / 2;
  const exEdge = addEdges(ex, accentColor, 0.8);
  registerPart(exEdge, new THREE.Vector3(0, 0, -1));
}

// ■■ ARC REACTOR ■■■
```

```
  registerPart(coreEdge, new THREE.Vector3(0, 0, 0));

  // Housing disc
  const houseGeo = new THREE.CylinderGeometry(1.0, 1.0, 0.16, 32, 1, true);
  const house = new THREE.Mesh(houseGeo, holoMat(secondaryColor, 0.10));
  house.rotation.x = Math.PI / 2;
  const houseEdge = addEdges(house, secondaryColor, 0.3);
  registerPart(houseEdge, new THREE.Vector3(0, 0, -0.5));
}

// ■■ DRONE / QUADCOPTER ■■■
```

```
    const legGeo = new THREE.BufferGeometry().setFromPoints(legPts);
    const leg = new THREE.Line(legGeo, lineMat(primaryColor, 0.5));
    registerPart(leg, new THREE.Vector3(0, -0.5, 0));
  });
}

// ■■ ROCKET ■■■
```

```
    registerPart(eEdge, new THREE.Vector3(Math.cos(a) * 0.3, -1.0, Math.sin(a) * 0.3));
  }
}

// ■■ V8 ENGINE ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
function buildV8Engine() {
  // Engine block (V configuration)
  [-0.28, 0.28].forEach((angle, side) => {
    // 4 cylinder heads per bank
    for (let i = 0; i < 4; i++) {
      const zOff = (i - 1.5) * 0.38;
      // Cylinder head
      const headGeo = new THREE.BoxGeometry(0.28, 0.18, 0.32);
      const head = new THREE.Mesh(headGeo, holoMat(primaryColor, 0.15));
      head.position.set(angle * 1.5, 0.35, zOff);
      head.rotation.z = angle * 0.6;
      const headEdge = addEdges(head, primaryColor, 0.8);
      headEdge.position.copy(head.position);
      headEdge.rotation.copy(head.rotation);
      registerPart(headEdge, new THREE.Vector3(angle * 1.2, 0.8, zOff * 0.5));

      // Piston (oscillating cylinder)
      const pisGeo = new THREE.CylinderGeometry(0.10, 0.10, 0.22, 20);
      const piston = new THREE.Mesh(pisGeo, holoMat(accentColor, 0.45));
      piston.position.set(angle * 0.6, -0.05, zOff);
      piston.rotation.z = angle * 0.8;
      const pisEdge = addEdges(piston, accentColor, 0.85);
      pisEdge.position.copy(piston.position);
      pisEdge.rotation.copy(piston.rotation);
      registerPart(pisEdge, new THREE.Vector3(angle, 0.4, zOff * 0.3));
    }
  });

  // Crankshaft
  const cshaftGeo = new THREE.CylinderGeometry(0.055, 0.055, 1.6, 20);
  const cshaft = new THREE.Mesh(cshaftGeo, holoMat(secondaryColor, 0.5));
  cshaft.rotation.x = Math.PI / 2;
  cshaft.position.set(0, -0.55, 0);
  const cshaftEdge = addEdges(cshaft, secondaryColor, 0.9);
  cshaftEdge.position.copy(cshaft.position);
  cshaftEdge.rotation.copy(cshaft.rotation);
  registerPart(cshaftEdge, new THREE.Vector3(0, -0.8, 0));

  // Oil pan
  const panGeo = new THREE.BoxGeometry(0.9, 0.14, 1.7);
  const pan = new THREE.Mesh(panGeo, holoMat(secondaryColor, 0.10));
  pan.position.set(0, -0.72, 0);
  const panEdge = addEdges(pan, secondaryColor, 0.5);
  panEdge.position.copy(pan.position);
  registerPart(panEdge, new THREE.Vector3(0, -1, 0));
}

// ■■ QUANTUM CPU ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
function buildQuantumCPU() {
  // Silicon die
  const dieGeo = new THREE.BoxGeometry(1.0, 0.06, 1.0);
  const die = new THREE.Mesh(dieGeo, holoMat(primaryColor, 0.15));
  const dieEdge = addEdges(die, primaryColor, 0.9);
  registerPart(dieEdge, new THREE.Vector3(0, 0, 0));

  // Qubit lattice (8x8 dots)
  for (let x = 0; x < 8; x++) {
    for (let z = 0; z < 8; z++) {
      const px = (x - 3.5) * 0.12;
      const pz = (z - 3.5) * 0.12;
      const qGeo = new THREE.SphereGeometry(0.025, 8, 8);
      const q = new THREE.Mesh(qGeo, holoMat(accentColor, 0.9));
      q.position.set(px, 0.05, pz);
```


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```
      registerPart(q, new THREE.Vector3(px * 0.5, 0.6, pz * 0.5));
    }
  }

  // Connecting lattice wires
  for (let x = 0; x < 7; x++) {
    for (let z = 0; z < 7; z++) {
      const px = (x - 3.5) * 0.12;
      const pz = (z - 3.5) * 0.12;
      const pts1 = [
        new THREE.Vector3(px, 0.05, pz),
        new THREE.Vector3(px + 0.12, 0.05, pz),
      ];
      const pts2 = [
        new THREE.Vector3(px, 0.05, pz),
        new THREE.Vector3(px, 0.05, pz + 0.12),
      ];
      [pts1, pts2].forEach(pts => {
        const wGeo = new THREE.BufferGeometry().setFromPoints(pts);
        const wire = new THREE.Line(wGeo, lineMat(primaryColor, 0.25));
        registerPart(wire, new THREE.Vector3(0, 0.3, 0));
      });
    }
  }

  // Cryo-coil rings (dilution refrigerator)
  [0.5, 0.75, 1.0].forEach((r, i) => {
    const torusGeo = new THREE.TorusGeometry(r, 0.03, 8, 48);
    const torus = new THREE.Mesh(torusGeo, holoMat(secondaryColor, 0.35));
    torus.position.set(0, -(i + 1) * 0.22, 0);
    const torusEdge = addEdges(torus, secondaryColor, 0.6);
    torusEdge.position.copy(torus.position);
    registerPart(torusEdge, new THREE.Vector3(0, -(i + 1) * 0.5, 0));
  });

  // Gold wire bonds (16 around perimeter)
  for (let i = 0; i < 16; i++) {
    const a = (i / 16) * Math.PI * 2;
    const pts = [
      new THREE.Vector3(Math.cos(a) * 0.45, 0.05, Math.sin(a) * 0.45),
      new THREE.Vector3(Math.cos(a) * 0.52, 0.15, Math.sin(a) * 0.52),
      new THREE.Vector3(Math.cos(a) * 0.54, 0.05, Math.sin(a) * 0.54),
    ];
    const wGeo = new THREE.BufferGeometry().setFromPoints(pts);
    const wire = new THREE.Line(wGeo, lineMat(0xffd600, 0.7));
    registerPart(wire, new THREE.Vector3(Math.cos(a) * 0.5, 0.4, Math.sin(a) * 0.5));
  }
}

// ■■ CYBERNETIC HEART ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
function buildCyberneticHeart() {
  // Left ventricle
  const lvGeo = new THREE.SphereGeometry(0.42, 24, 24);
  const lv = new THREE.Mesh(lvGeo, holoMat(primaryColor, 0.14));
  lv.position.set(-0.22, 0, 0);
  lv.scale.set(1, 1.2, 0.85);
  const lvEdge = addEdges(lv, primaryColor, 0.85);
  lvEdge.position.copy(lv.position);
  lvEdge.scale.copy(lv.scale);
  registerPart(lvEdge, new THREE.Vector3(-1, 0.3, 0));

  // Right ventricle
  const rvGeo = new THREE.SphereGeometry(0.35, 24, 24);
  const rv = new THREE.Mesh(rvGeo, holoMat(accentColor, 0.14));
  rv.position.set(0.22, 0.1, 0);
  rv.scale.set(1, 1.1, 0.8);
  const rvEdge = addEdges(rv, accentColor, 0.8);
  rvEdge.position.copy(rv.position);
```


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```
  rvEdge.scale.copy(rv.scale);
  registerPart(rvEdge, new THREE.Vector3(1, 0.3, 0));

  // Aorta arch
  const aortaPts = [];
  for (let t = 0; t <= 32; t++) {
    const a = (t / 32) * Math.PI;
    aortaPts.push(new THREE.Vector3(
      -0.22 + Math.cos(a) * 0.35,
      0.45 + Math.sin(a) * 0.28,
      0
    ));
  }
  const aortaGeo = new THREE.BufferGeometry().setFromPoints(aortaPts);
  const aorta = new THREE.Line(aortaGeo, lineMat(secondaryColor, 0.9));
  registerPart(aorta, new THREE.Vector3(0, 1, 0));

  // Pulmonary artery
  const paPts = [
    new THREE.Vector3(0.22, 0.4, 0),
    new THREE.Vector3(0.22, 0.7, 0),
    new THREE.Vector3(0.0, 0.8, 0),
  ];
  const paGeo = new THREE.BufferGeometry().setFromPoints(paPts);
  const pa = new THREE.Line(paGeo, lineMat(accentColor, 0.8));
  registerPart(pa, new THREE.Vector3(0.5, 0.8, 0));

  // Valve rings (4)
  const valvePts = [
    { pos: [-0.22, 0, 0], r: 0.15 },
    { pos: [ 0.22, 0, 0], r: 0.12 },
    { pos: [-0.22, 0.45, 0], r: 0.10 },
    { pos: [ 0.22, 0.40, 0], r: 0.09 },
  ];
  valvePts.forEach(({ pos, r }, i) => {
    const vGeo = new THREE.TorusGeometry(r, 0.018, 8, 32);
    const valve = new THREE.Mesh(vGeo, holoMat(0xffffff, 0.55));
    valve.position.set(...pos);
    const vEdge = addEdges(valve, 0xffffff, 0.85);
    vEdge.position.copy(valve.position);
    registerPart(vEdge, new THREE.Vector3(pos[0] * 0.8, 0.6, 0));
  });
}

// ■■ IRON MAN HELMET ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
function buildIronManHelmet() {
  // Main dome
  const domeGeo = new THREE.SphereGeometry(0.7, 32, 24, 0, Math.PI * 2, 0, Math.PI * 0.6);
  const dome = new THREE.Mesh(domeGeo, holoMat(primaryColor, 0.12));
  dome.position.set(0, 0.1, 0);
  const domeEdge = addEdges(dome, primaryColor, 0.8);
  domeEdge.position.copy(dome.position);
  registerPart(domeEdge, new THREE.Vector3(0, 1, 0));

  // Faceplate
  const faceGeo = new THREE.SphereGeometry(0.65, 32, 24, 0, Math.PI * 2, Math.PI * 0.35, Math.PI * 0.35);
  const face = new THREE.Mesh(faceGeo, holoMat(accentColor, 0.15));
  const faceEdge = addEdges(face, accentColor, 0.85);
  registerPart(faceEdge, new THREE.Vector3(0, -0.2, 1));

  // Eye slots (2 rectangular openings)
  [-0.22, 0.22].forEach(x => {
    const eyeGeo = new THREE.BoxGeometry(0.18, 0.07, 0.35);
    const eye = new THREE.Mesh(eyeGeo, holoMat(0xffd600, 0.9));
    eye.position.set(x, 0.15, 0.55);
    const eyeEdge = addEdges(eye, 0xffd600, 1.0);
    eyeEdge.position.copy(eye.position);
    registerPart(eyeEdge, new THREE.Vector3(x * 2, 0.3, 1.5));
```


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```
  });

  // Chin guard
  const chinGeo = new THREE.BoxGeometry(0.72, 0.20, 0.60);
  const chin = new THREE.Mesh(chinGeo, holoMat(primaryColor, 0.14));
  chin.position.set(0, -0.52, 0.22);
  chin.scale.z = 0.8;
  const chinEdge = addEdges(chin, primaryColor, 0.75);
  chinEdge.position.copy(chin.position);
  chinEdge.scale.copy(chin.scale);
  registerPart(chinEdge, new THREE.Vector3(0, -1, 0.5));

  // Side panels
  [-0.68, 0.68].forEach(x => {
    const panGeo = new THREE.BoxGeometry(0.12, 0.55, 0.55);
    const panel = new THREE.Mesh(panGeo, holoMat(secondaryColor, 0.18));
    panel.position.set(x, 0.0, 0.15);
    const panEdge = addEdges(panel, secondaryColor, 0.75);
    panEdge.position.copy(panel.position);
    registerPart(panEdge, new THREE.Vector3(x * 1.5, 0, 0.5));
  });
}

// ■■ SATELLITE ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
function buildSatellite() {
  // Main satellite body
  const bodyGeo = new THREE.BoxGeometry(0.5, 0.35, 0.65);
  const body = new THREE.Mesh(bodyGeo, holoMat(primaryColor, 0.15));
  const bodyEdge = addEdges(body, primaryColor, 0.85);
  registerPart(bodyEdge, new THREE.Vector3(0, 0.3, 0));

  // 4 Solar panels (2 wings × 2 panels each)
  [[-1.8, 0, 0], [1.8, 0, 0]].forEach((pos, wingIdx) => {
    for (let p = 0; p < 2; p++) {
      const panGeo = new THREE.BoxGeometry(0.9, 0.02, 0.38);
      const pan = new THREE.Mesh(panGeo, holoMat(accentColor, 0.20));
      pan.position.set(pos[0] + (p === 0 ? -0.55 : 0.55), 0, 0);
      const panEdge = addEdges(pan, accentColor, 0.8);
      panEdge.position.copy(pan.position);
      registerPart(panEdge, new THREE.Vector3(pos[0] * 1.3, 0, 0));
    }
    // Panel arm
    const armPts = [new THREE.Vector3(0, 0, 0), new THREE.Vector3(pos[0], 0, 0)];
    const armGeo = new THREE.BufferGeometry().setFromPoints(armPts);
    const arm = new THREE.Line(armGeo, lineMat(primaryColor, 0.7));
    registerPart(arm, new THREE.Vector3(pos[0] * 0.5, 0.3, 0));
  });

  // Phased array antenna (dish approximation)
  const dishGeo = new THREE.CylinderGeometry(0.22, 0.01, 0.14, 32, 1, true);
  const dish = new THREE.Mesh(dishGeo, holoMat(secondaryColor, 0.18));
  dish.position.set(0, 0.25, 0);
  const dishEdge = addEdges(dish, secondaryColor, 0.9);
  dishEdge.position.copy(dish.position);
  registerPart(dishEdge, new THREE.Vector3(0, 1, 0));

  // Thruster nozzles (2)
  [-0.15, 0.15].forEach(x => {
    const nGeo = new THREE.CylinderGeometry(0.04, 0.07, 0.14, 16);
    const nozzle = new THREE.Mesh(nGeo, holoMat(accentColor, 0.55));
    nozzle.position.set(x, -0.25, 0);
    const nEdge = addEdges(nozzle, accentColor, 0.85);
    nEdge.position.copy(nozzle.position);
    registerPart(nEdge, new THREE.Vector3(x, -0.8, 0));
  });
}

// ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
```


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```
// SCHEMATIC SWITCHER (called when Python polls)
// ■■■
```

```
      } else if (shape === 'cylinder') {
        geo = new THREE.CylinderGeometry(p[0] || 0.5, p[1] || 0.5, p[2] || 1, p[3] || 24);
      } else if (shape === 'sphere') {
        geo = new THREE.SphereGeometry(p[0] || 0.5, p[1] || 24, p[2] || 24);
      } else if (shape === 'cone') {
        geo = new THREE.ConeGeometry(p[0] || 0.5, p[1] || 1, p[2] || 24);
      } else if (shape === 'torus') {
        geo = new THREE.TorusGeometry(p[0] || 0.8, p[1] || 0.08, p[2] || 12, p[3] || 32);
      } else {
        geo = new THREE.BoxGeometry(p[0] || 1, p[1] || 1, p[2] || 1);
      }
    } catch(e) {
      geo = new THREE.BoxGeometry(1, 1, 1);
    }

    const c = part.color ? parseInt(part.color.replace('#',''), 16) : primaryColor;
    const op = part.opacity !== undefined ? part.opacity : 0.35;
    const mesh = new THREE.Mesh(geo, holoMat(c, op));

    if (part.position) mesh.position.set(part.position[0], part.position[1], part.position[2]);
    if (part.rotation) mesh.rotation.set(part.rotation[0], part.rotation[1], part.rotation[2]);
    if (part.scale) mesh.scale.set(part.scale[0], part.scale[1], part.scale[2]);

    const edge = addEdges(mesh, c, Math.min(1.0, op + 0.45));
    edge.position.copy(mesh.position);
    edge.rotation.copy(mesh.rotation);
    edge.scale.copy(mesh.scale);

    const dir = part.explode_dir
      ? new THREE.Vector3(part.explode_dir[0], part.explode_dir[1], part.explode_dir[2]).normalize()
      : (mesh.position.length() > 0.01 ? mesh.position.clone().normalize() : new THREE.Vector3(0, 1, 0));

    registerPart(mesh, dir);
    registerPart(edge, dir);
  });
}

// ═══════════════════════════════════════════════════════════════
// CONTROLS
// ═══════════════════════════════════════════════════════════════
function toggleExplode() {
  exploded = !exploded;
  document.getElementById('btn-explode').classList.toggle('active', exploded);
}

function toggleSpin() {
  spinning = !spinning;
  controls.autoRotate = spinning;
  document.getElementById('btn-spin').classList.toggle('active', spinning);
}

function toggleXray() {
  xrayMode = !xrayMode;
  document.getElementById('btn-xray').classList.toggle('active', xrayMode);
  modelParts.forEach(p => {
    if (p.mesh.material) {
      p.mesh.material.wireframe = xrayMode;
      p.mesh.material.opacity = xrayMode ? 0.3 : (p.mesh.material._baseOpacity || 0.85);
    }
  });
}

// ═══════════════════════════════════════════════════════════════
// DATA POLLING
// ═══════════════════════════════════════════════════════════════
let _lastSchematicKey = '';

async function pollSchematic() {
```


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```
  try {
    const r = await fetch('/api/schematic');
    if (!r.ok) return;
    const data = await r.json();
    if (!data || !data.id) return;
    const key = data.id;
    if (key !== _lastSchematicKey) {
      _lastSchematicKey = key;
      loadSchematic(data);
    }
  } catch(e) {}
}

setInterval(pollSchematic, 1200);

// ═══════════════════════════════════════════════════════════════════════════
// ANIMATION LOOP
// ═══════════════════════════════════════════════════════════════════════════
let t = 0;

function animate() {
  requestAnimationFrame(animate);
  t += 0.016;

  // OrbitControls handles smooth 360-degree rotation and mouse damping
  controls.update();

  // Explode / contract parts
  explodeT = exploded
    ? Math.min(1, explodeT + 0.03)
    : Math.max(0, explodeT - 0.03);

  modelParts.forEach(p => {
    const ease = explodeT * explodeT * (3 - 2 * explodeT); // smoothstep
    p.mesh.position.lerpVectors(p.originPos, p.originPos.clone().add(p.explodeDir.clone().multiplyScalar(1.4)),
ease);
  });

  // Pulse HUD rings
  ring1.rotation.y = t * 0.3;
  ring2.rotation.y = -t * 0.2;

  // Particle drift
  const pPos = particles.geometry.attributes.position;
  for (let i = 0; i < pPos.count; i++) {
    let py = pPos.getY(i);
    py += 0.003;
    if (py > 5) py -= 10;
    pPos.setY(i, py);
  }
  pPos.needsUpdate = true;

  renderer.render(scene, camera);
}

// ═══════════════════════════════════════════════════════════════════════════
// WINDOW RESIZE
// ═══════════════════════════════════════════════════════════════════════════
window.addEventListener('resize', () => {
  const w = window.innerWidth;
  const h = window.innerHeight - 36;
  renderer.setSize(w, h);
  camera.aspect = w / h;
  camera.updateProjectionMatrix();
});

// ═══════════════════════════════════════════════════════════════════════════
// BOOT
```

```
// ═══════════════════════════════════════════════════════════════════════════
function initMatrix() {
  pollSchematic();
  animate();
  setTimeout(() => {
    if (currentSchematicId === '') {
      buildTurbofan();
      document.getElementById('spec-name').textContent = 'CFM LEAP-1B Turbofan Engine';
    }
    const ldr = document.getElementById('loader');
    if (ldr) ldr.classList.add('gone');
  }, 600);
}

if (document.readyState === 'complete' || document.readyState === 'interactive') {
  initMatrix();
} else {
  window.addEventListener('DOMContentLoaded', initMatrix);
}
</script>
</body>
</html>"""

# ═══════════════════════════════════════════════════════════════════════════
# FLASK SERVER
# ═══════════════════════════════════════════════════════════════════════════

def _make_app():
    app = Flask(__name__)

    # Serve Three.js and OrbitControls from local files (no CDN dependency)
    _threejs_path = os.path.join(os.path.dirname(os.path.abspath(__file__)), "three.min.js")
    _orbit_path = os.path.join(os.path.dirname(os.path.abspath(__file__)), "OrbitControls.js")

    @app.route("/three.js")
    def serve_threejs():
        if os.path.exists(_threejs_path):
            with open(_threejs_path, "rb") as f:
                content = f.read()
            from flask import Response
            return Response(content, mimetype="application/javascript")
        return "// Three.js not found", 404

    @app.route("/OrbitControls.js")
    def serve_orbitcontrols():
        if os.path.exists(_orbit_path):
            with open(_orbit_path, "rb") as f:
                content = f.read()
            from flask import Response
            return Response(content, mimetype="application/javascript")
        return "// OrbitControls not found", 404

    @app.route("/")
    def index():
        return render_template_string(_HTML)

    @app.route("/api/schematic")
    def get_schematic():
        return jsonify(_current_schematic)

    @app.route("/api/set_schematic", methods=["POST"])
    def set_schematic():
        global _current_schematic
        _current_schematic = _flask_req.get_json(force=True) or {}
        return jsonify({"ok": True})

    return app
```

```
def _run_server():
    global _server_started
    if not _HAS_FLASK:
        return
    logging.getLogger("werkzeug").setLevel(logging.ERROR)
    _server_started = True
    _make_app().run(
        host="127.0.0.1",
        port=FLASK_PORT,
        debug=False,
        use_reloader=False,
        threaded=True,
    )


def _ensure_server():
    global _flask_thread, _server_started
    if _server_started:
        return
    if not _HAS_FLASK:
        return
    _flask_thread = threading.Thread(target=_run_server, daemon=True)
    _flask_thread.start()
    deadline = time.time() + 6.0
    while time.time() < deadline:
        try:
            _requests.get(f"http://127.0.0.1:{FLASK_PORT}/api/schematic", timeout=0.4)
            break
        except Exception:
            time.sleep(0.1)


def _open_viewer(payload: dict):
    """Ensure server is running, push payload, and open the transparent viewer window."""
    global _viewer_process

    # Step 1: Start Flask and block until it is actually accepting connections
    _ensure_server()

    # Step 2: Push schematic payload BEFORE opening the viewer (retry up to 5 times)
    if payload and _HAS_REQUESTS and _server_started:
        for attempt in range(5):
            try:
                resp = _requests.post(
                    f"http://127.0.0.1:{FLASK_PORT}/api/set_schematic",
                    json=payload,
                    timeout=2,
                )
                if resp.ok:
                    break
            except Exception:
                time.sleep(0.3)

    viewer_py = os.path.join(os.path.dirname(os.path.abspath(__file__)), "schematic_viewer.py")

    if _HAS_WEBVIEW and os.path.exists(viewer_py):
        # Kill any existing viewer process first
        if _viewer_process and _viewer_process.poll() is None:
            try:
                _viewer_process.terminate()
                time.sleep(0.3)
            except Exception:
                pass

        try:
            # Use pythonw.exe on Windows for clean GUI execution without black console
            py_exe = os.path.join(os.path.dirname(sys.executable), "pythonw.exe")
```

```
            if not os.path.exists(py_exe):
                py_exe = sys.executable

            _viewer_process = subprocess.Popen(
                [py_exe, viewer_py, str(FLASK_PORT)],
            )
            time.sleep(1.0)
        except Exception:
            import webbrowser
            webbrowser.open(f"http://127.0.0.1:{FLASK_PORT}/")
    else:
        import webbrowser
        webbrowser.open(f"http://127.0.0.1:{FLASK_PORT}/")



def _push_schematic(payload: dict):
    """Push a new schematic to an already-running viewer (no new window)."""
    _ensure_server()
    if _HAS_REQUESTS and _server_started:
        try:
            _requests.post(
                f"http://127.0.0.1:{FLASK_PORT}/api/set_schematic",
                json=payload,
                timeout=2,
            )
        except Exception:
            pass


# ■■■
```

```
def dismiss_schematic() -> str:
    """
    JARVIS TOOL — Dismiss holographic schematic.
    Closes the floating 3D blueprint hologram window.

    Returns:
        Status string for JARVIS to speak.
    """
    global _viewer_process, _current_schematic
    _current_schematic = {}

    if _viewer_process is not None:
        try:
            _viewer_process.terminate()
            _viewer_process = None
            return "SUCCESS: Holographic schematic dismissed, sir."
        except Exception as exc:
            return f"ERROR: Could not dismiss schematic — {exc}"

    return "SUCCESS: No active schematic to dismiss, sir."


# ═══════════════════════════════════════════════════════════════════════════
# STANDALONE ENTRYPOINT
# ═══════════════════════════════════════════════════════════════════════════
if __name__ == "__main__":
    import sys as _sys
    _sys.stdout.reconfigure(encoding="utf-8", errors="replace")
    query_arg = " ".join(_sys.argv[1:]) if len(_sys.argv) > 1 else "airplane engine"
    result = launch_schematic_hud(query_arg, auto_open=True)
    print(result)
    try:
        while True:
            time.sleep(1)
    except KeyboardInterrupt:
        pass
```

// MODULE: schematic_viewer.py (PYWEBVIEW TRANSPARENT DESKTOP WIDGET LAUNCHER)

```
"""
schematic_viewer.py
Project Schematic — JARVIS Holographic Blueprint System
Dedicated subprocess launcher so pywebview runs on its own main thread.
Transparent, frameless window for true hologram-floating-over-desktop effect.
Built by Swayam Dhawale, Pulse of AI.

Usage: python schematic_viewer.py [port]
"""
import sys
import webview


def main():
    port = int(sys.argv[1]) if len(sys.argv) > 1 else 9393
    url = f"http://127.0.0.1:{port}/"

    window = webview.create_window(
        title="JARVIS :: PROJECT SCHEMATIC",
        url=url,
        width=1100,
        height=800,
        resizable=True,
        frameless=True,
        transparent=False,
        on_top=True,
        background_color="#000608",
```

```
        text_select=False,
        min_size=(700, 550),
    )
    webview.start(debug=False, http_server=False, gui="edgechromium")


if __name__ == "__main__":
    main()
```

// MODULE: agent.py (CORE GEMINI LIVE ENGINE (UPDATED WITH SCHEMATIC TOOL))

```
"""
agent.py
Real-Time Gemini Live Engine with Semi-Transparent Cyberpunk HUD Console (Episode 23)
Autonomous App Orchestration, Multi-Tab Web Swarm, PPT/PDF Engine & Master Serious Mode Pipeline
Built by Swayam Dhawale, Founder of Pulse of AI.
"""

import asyncio
import datetime
import os
import sys
import time
import math
import struct
import threading
import subprocess
import collections
import queue
import tkinter as tk
from tkinter import scrolledtext
import pyaudio
import winsound
import keyboard
import requests
from dotenv import load_dotenv
from google import genai
from google.genai import errors, types
import websockets
from rich.console import Console
import fui_bridge

load_dotenv()
if sys.platform == "win32":
    try:
        sys.stdout.reconfigure(encoding="utf-8")
        sys.stderr.reconfigure(encoding="utf-8")
    except Exception:
        pass
console = Console()

# --- REMOTE KILL-SWITCH (Pulse of AI) ---
STATUS_URL = "https://gist.githubusercontent.com/PulseOfAIDev/a9951c6e858aacbabb604a0ed23f6ab7/raw/status.json"

def verify_app_access():
    """Checks remote status from GitHub Gist and displays cloud-configured message."""
    try:
        fresh_url = f"{STATUS_URL}?t={int(time.time())}"
        response = requests.get(fresh_url, headers={"Cache-Control": "no-cache"}, timeout=4).json()

        status = str(response.get("app_status", "free")).strip().lower()
        message = response.get(
            "message",
            "The free beta access period has concluded. To activate a subscription tier or upgrade your account,
visit: https://pulseofaidev.github.io"
        )
```

```
        if status != "free":
            console.print(f"\n[bold red]■ [ACCESS RESTRICTED][/bold red] {message}\n")
            sys.exit(0)
    except Exception:
        pass

# Run verification immediately on boot
verify_app_access()

# --- SECURE KEYRING INTEGRATION ---
from Setup import get_or_prompt_api_key

# Hardware & Telemetry
from hardware_control import (
    get_brightness,
    get_volume,
    set_brightness,
    set_volume,
    get_battery_status,
    get_system_vitals,
    get_system_temperatures,
    restore_master_volume,
)

# Software, Media Telemetry & Navigation Tools
from software_control import (
    open_website,
    open_application,
    close_application,
    purge_background_tasks,
    control_media,
    get_current_media_info,
    is_media_actively_playing,
    navigate_shorts,
    skip_video,
    seek_video,
    toggle_video_view,
)

# Local Zero-Latency Vision & Autonomous Grounding Tools
from screen_vision import (
    analyze_screen,
    find_ui_element,
    skip_youtube_ad,
)

# Self-Modifying Tri-Core Dynamic Skill Synthesizer
from skill_synthesizer import (
    evolve_skill,
    list_active_skills,
)

# Fast Cloud Reasoning Tool
from llm_router import (
    query_deepseek_reasoning,
)

# Episode 17: Autonomous App Composer & Typing Engine
from app_automator import (
    open_and_compose,
)

# Episode 18 & 19: Autonomous PowerPoint Generator & Presentation Engine
from ppt_engine import (
    create_and_launch_ppt,
)
```

```
# Episode 19 & 20: Autonomous Executive PDF Briefing Engine & Episode 23 Master Pipeline
from pdf_engine import (
    research_and_generate_pdf,
    execute_serious_mode,
)

# Episode 21: Autonomous Web Swarm & Browser Automation
from web_crawler import (
    run_autonomous_swarm,
)

# Episode 22: Local GPU Offline RAG Brain
from local_rag_engine import (
    process_dossier_offline,
)

# Episode 26: Autonomous Optical Physical Vision & Sentry Defense Core
from sentry_vision import (
    inspect_physical_world,
    activate_sentry_mode,
    deactivate_sentry_mode,
    get_sentry_security_report,
    capture_camera_snapshot,
    check_user_presence,
)

# Episode 27: Autonomous Screen Auto-Clicker & GUI Grounding Engine
from screen_clicker import (
    click_on_screen,
    double_click_on_screen,
    right_click_on_screen,
    move_cursor_to_element,
)

# Episode 28: Autonomous Desktop Pilot & Computer-Use Operator (Mark 85 Omni-Pilot)
from desktop_pilot import (
    execute_autonomous_desktop_pilot,
    abort_desktop_pilot,
)

# Episode 29: Autonomous Ambient Ghost Operator & Executive Shadow Radar (Project Doppelgänger)
from ghost_operator import (
    activate_ghost_mode,
    deactivate_ghost_mode,
    get_ghost_radar_status,
    prepare_meeting_briefing,
    apply_ghost_recommendation,
    analyze_clipboard_context,
    ghost_operator,
)

# Episode 30: 100% Local LLM & Autonomous Offline Sovereign Core (Project Sovereign)
from offline_engine import (
    offline_jarvis,
    activate_offline_mode,
    deactivate_offline_mode,
    get_offline_system_status,
    check_internet_connection,
)

# Project Orbital: 3D Holographic Earth HUD & Live Telemetry Grid
from orbital_hud import (
    launch_orbital_hud,
)

# Project Sentinel: Advanced 3D Globe HUD with Trajectory Arc Engine
from globe_hud import (
    launch_globe_hud,
```

```
)

# Project Schematic: Floating Transparent 3D Holographic Blueprint System
from schematic_hud import (
    launch_schematic_hud,
    dismiss_schematic,
)

# Dynamically initialize Gemini client using the secure keyring setup
client = genai.Client(api_key=get_or_prompt_api_key())

FORMAT = pyaudio.paInt16
CHANNELS = 1
RATE_IN = 16000
RATE_OUT = 24000
CHUNK = 512

p = pyaudio.PyAudio()

model_is_speaking = False
server_turn_active = False
is_user_speaking = False
discard_current_turn = False
acoustic_barge_in_enabled = True
media_is_playing_state = False
audio_player = None

# Dynamic Auto-Calibrated Audio Thresholds (tuned for natural voice sensitivity)
ambient_noise_p95 = 350.0
speech_start_threshold = 1100.0
speech_continue_threshold = 650.0
barge_in_threshold = 3800.0

class SmoothAudioPlayer:
    """Zero-latency, jitter-buffered background audio playback worker.
    Eliminates underflows, stuttering, and robotic audio lag on Windows.
    """
    def __init__(self, sample_rate=24000, buffer_frames=1024):
        self.sample_rate = sample_rate
        self.buffer_frames = buffer_frames
        self.audio_queue = queue.Queue()
        self.is_playing = False
        self.stop_flag = threading.Event()
        self.last_play_time = 0.0
        self.thread = threading.Thread(target=self._play_loop, daemon=True)
        self.thread.start()

    def _play_loop(self):
        try:
            stream = p.open(
                format=FORMAT,
                channels=CHANNELS,
                rate=self.sample_rate,
                output=True,
                frames_per_buffer=self.buffer_frames,
            )
        except Exception as e:
            console.print(f"[red]Error opening audio output stream: {e}[/red]")
            return

        try:
            while not self.stop_flag.is_set():
                try:
                    chunk = self.audio_queue.get(timeout=0.05)
                except queue.Empty:
                    if self.is_playing:
                        self.is_playing = False
                        self.last_play_time = time.monotonic()
```

```
                    continue

                self.is_playing = True

                # Pre-buffering to absorb WebSocket jitter on the first packet of a sentence
                chunks = [chunk]
                for _ in range(3):
                    try:
                        chunks.append(self.audio_queue.get_nowait())
                    except queue.Empty:
                        break

                initial_burst = b"".join(chunks)
                try:
                    stream.write(initial_burst)
                    self.last_play_time = time.monotonic()
                    energy = calculate_rms(initial_burst)
                    fui_bridge.broadcast_audio_energy(min(1.0, energy / 2500.0), "jarvis")
                except Exception:
                    pass

                # Stream continuous audio smoothly without underflows
                while not self.stop_flag.is_set():
                    try:
                        chunk = self.audio_queue.get(timeout=0.15)
                        stream.write(chunk)
                        self.last_play_time = time.monotonic()
                        try:
                            energy = calculate_rms(chunk)
                            fui_bridge.broadcast_audio_energy(min(1.0, energy / 2500.0), "jarvis")
                        except Exception:
                            pass
                    except queue.Empty:
                        break

                self.is_playing = False
                self.last_play_time = time.monotonic()
            finally:
                try:
                    stream.stop_stream()
                    stream.close()
                except Exception:
                    pass

    def feed(self, data: bytes):
        if data:
            self.audio_queue.put(data)

    def interrupt(self):
        """Instantly flushes pending audio chunks and silences speaker."""
        while not self.audio_queue.empty():
            try:
                self.audio_queue.get_nowait()
            except queue.Empty:
                break
        self.is_playing = False

    def close(self):
        self.stop_flag.set()
        self.interrupt()
        self.thread.join(timeout=1.0)


# --- Semi-Transparent Cyberpunk Neural HUD with Mini-Mode ---
class JarvisHUD:
    def __init__(self, send_callback):
        self.send_callback = send_callback
        self.is_compact = False
```


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```
self.root = tk.Tk()
self.root.title("■ JARVIS NEURAL CONSOLE")
self.root.geometry("640x520")
self.root.configure(bg="#0B0F19")

# Cyberpunk floating overlay styling
self.root.attributes("-topmost", True)
self.root.attributes("-alpha", 0.88)

# Header Bar Frame (Houses Title & Compact Switch)
self.header_frame = tk.Frame(self.root, bg="#0B0F19")
self.header_frame.pack(fill=tk.X, padx=12, pady=(6, 2))

self.header_label = tk.Label(
    self.header_frame,
    text="JARVIS NEURAL CONSOLE",
    font=("Consolas", 11, "bold"),
    fg="#00E5FF",
    bg="#0B0F19"
)
self.header_label.pack(side=tk.LEFT)

self.toggle_btn = tk.Button(
    self.header_frame,
    text="[—] COMPACT",
    font=("Consolas", 8, "bold"),
    bg="#1E293B",
    fg="#38BDF8",
    activebackground="#0369A1",
    activeforeground="#FFFFFF",
    relief=tk.FLAT,
    padx=6,
    command=self.toggle_mode
)
self.toggle_btn.pack(side=tk.RIGHT)

self.sub_header = tk.Label(
    self.root,
    text="Voice Engine: Gemini Live | Research Swarm & Deep Reasoning: Groq / OpenRouter / Ollama",
    font=("Consolas", 8),
    fg="#70829B",
    bg="#0B0F19"
)
self.sub_header.pack(pady=(0, 4))

# Translucent Chat Log Display
self.chat_display = scrolledtext.ScrolledText(
    self.root,
    wrap=tk.WORD,
    font=("Consolas", 10),
    bg="#111827",
    fg="#E2E8F0",
    insertbackground="#00E5FF",
    relief=tk.FLAT,
    padx=10,
    pady=8
)
self.chat_display.pack(padx=12, pady=4, fill=tk.BOTH, expand=True)
self.chat_display.config(state=tk.DISABLED)

# Dialogue & status tags
self.chat_display.tag_config("user_voice", foreground="#38BDF8", font=("Consolas", 10, "bold"))
self.chat_display.tag_config("user_text", foreground="#A78BFA", font=("Consolas", 10, "bold"))
self.chat_display.tag_config("jarvis_header", foreground="#4ADE80", font=("Consolas", 10, "bold"))
self.chat_display.tag_config("jarvis_stream", foreground="#86EFAC", font=("Consolas", 10))
self.chat_display.tag_config("tool", foreground="#FBBF24", font=("Consolas", 9, "italic"))
self.chat_display.tag_config("system", foreground="#94A3B8", font=("Consolas", 9))
```


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```
        # Keyboard Command Input Bar
        self.input_frame = tk.Frame(self.root, bg="#0B0F19")
        self.input_frame.pack(fill=tk.X, padx=12, pady=(4, 10))

        self.entry_box = tk.Entry(
            self.input_frame,
            font=("Consolas", 10),
            bg="#1E293B",
            fg="#F8FAFC",
            insertbackground="#00E5FF",
            relief=tk.FLAT
        )
        self.entry_box.pack(side=tk.LEFT, fill=tk.X, expand=True, ipady=6, padx=(0, 6))
        self.entry_box.bind("<Return>", self.on_send)

        self.send_btn = tk.Button(
            self.input_frame,
            text="TRANSMIT",
            font=("Consolas", 9, "bold"),
            bg="#0284C7",
            fg="#FFFFFF",
            activebackground="#0369A1",
            activeforeground="#FFFFFF",
            relief=tk.FLAT,
            padx=14,
            command=self.on_send
        )
        self.send_btn.pack(side=tk.RIGHT, ipady=4)

    def toggle_mode(self):
        """Switches between Full Log HUD and Sleek Floating Input Bar."""
        if not self.is_compact:
            self.chat_display.pack_forget()
            self.sub_header.pack_forget()
            self.root.geometry("450x85")
            self.toggle_btn.config(text="[+] EXPAND")
            self.is_compact = True
        else:
            self.root.geometry("640x520")
            self.sub_header.pack(pady=(0, 4), before=self.input_frame)
            self.chat_display.pack(padx=12, pady=4, fill=tk.BOTH, expand=True, before=self.input_frame)
            self.toggle_btn.config(text="[—] COMPACT")
            self.is_compact = False

    def enable_serious_theme(self):
        """Dynamically shifts HUD to Tactical Crimson when Serious Mode activates."""
        self.root.configure(bg="#1A0505")
        self.header_frame.configure(bg="#1A0505")
        self.header_label.configure(fg="#FF2A2A", bg="#1A0505", text="■ JARVIS [SERIOUS MODE ACTIVE]")
        self.sub_header.configure(fg="#FF7777", bg="#1A0505")
        self.chat_display.configure(bg="#2D0A0A", fg="#FFEBEB", insertbackground="#FF2A2A")
        self.input_frame.configure(bg="#1A0505")
        self.entry_box.configure(bg="#3D0F0F", fg="#FFFFFF", insertbackground="#FF2A2A")
        self.send_btn.configure(bg="#B91C1C", activebackground="#991B1B")
        self.toggle_btn.configure(bg="#3D0F0F", fg="#FF2A2A", activebackground="#7F1D1D")

    def restore_normal_theme(self):
        """Restores HUD to standard Cyberpunk Blue when Serious Mode completes."""
        self.root.configure(bg="#0B0F19")
        self.header_frame.configure(bg="#0B0F19")
        self.header_label.configure(fg="#00E5FF", bg="#0B0F19", text="JARVIS NEURAL CONSOLE")
        self.sub_header.configure(fg="#70829B", bg="#0B0F19")
        self.chat_display.configure(bg="#111827", fg="#E2E8F0", insertbackground="#00E5FF")
        self.input_frame.configure(bg="#0B0F19")
        self.entry_box.configure(bg="#1E293B", fg="#F8FAFC", insertbackground="#00E5FF")
        self.send_btn.configure(bg="#0284C7", activebackground="#0369A1")
        self.toggle_btn.configure(bg="#1E293B", fg="#38BDF8", activebackground="#0369A1")
```

```
    def append_log(self, text: str, tag: str = "system"):
        self.chat_display.config(state=tk.NORMAL)
        self.chat_display.insert(tk.END, text + "\n", tag)
        self.chat_display.see(tk.END)
        self.chat_display.config(state=tk.DISABLED)

    def append_stream_chunk(self, token: str, tag: str = "jarvis_stream"):
        self.chat_display.config(state=tk.NORMAL)
        self.chat_display.insert(tk.END, token, tag)
        self.chat_display.see(tk.END)
        self.chat_display.config(state=tk.DISABLED)

    def on_send(self, event=None):
        msg = self.entry_box.get().strip()
        if msg:
            self.append_log(f"YOU (HUD): {msg}", "user_text")
            self.entry_box.delete(0, tk.END)
            self.send_callback(msg)

    def run(self):
        self.root.mainloop()

hud_instance = None
fui_process = None

def start_gui(loop, text_queue):
    global hud_instance
    def on_user_submit(message: str):
        asyncio.run_coroutine_threadsafe(text_queue.put(message), loop)

    hud_instance = JarvisHUD(send_callback=on_user_submit)
    hud_instance.run()

def log_to_gui(text: str, tag: str = "system"):
    if hud_instance:
        hud_instance.root.after(0, hud_instance.append_log, text, tag)
    fui_bridge.broadcast_log(text, tag)

def stream_token_to_gui(token: str, tag: str = "jarvis_stream"):
    if hud_instance:
        hud_instance.root.after(0, hud_instance.append_stream_chunk, token, tag)
    fui_bridge.broadcast_token(token, tag)

def trigger_serious_hud():
    if hud_instance:
        hud_instance.root.after(0, hud_instance.enable_serious_theme)
    fui_bridge.set_fui_mode("serious")

def trigger_normal_hud():
    if hud_instance:
        hud_instance.root.after(0, hud_instance.restore_normal_theme)
    fui_bridge.set_fui_mode("normal")


def calculate_rms(data: bytes) -> float:
    """Calculates RMS audio energy."""
    count = len(data) // 2
    if count == 0:
        return 0.0
    shorts = struct.unpack(f"{count}h", data)
    sum_squares = sum(s * s for s in shorts)
    return math.sqrt(sum_squares / count)

def is_jarvis_speaking() -> bool:
    """Returns True if Gemini is sending voice, speakers are playing, or echo guard is active."""
    global server_turn_active, audio_player
    if server_turn_active:
```


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```
        return True
    if audio_player and audio_player.is_playing:
        return True
    # 400ms acoustic room reverb decay guard to prevent speaker bleed
    if audio_player and (time.monotonic() - audio_player.last_play_time < 0.40):
        return True
    return False

def calibrate_ambient_noise(mic_stream):
    """Dynamically samples ambient room noise to establish natural, sensitive VAD and responsive barge-in
thresholds."""
    global ambient_noise_p95, speech_start_threshold, speech_continue_threshold, barge_in_threshold
    try:
        # Discard initial driver startup transients and Enter key click
        for _ in range(8):
            mic_stream.read(CHUNK, exception_on_overflow=False)

        readings = []
        for _ in range(40): # ~1.3 seconds sample
            data = mic_stream.read(CHUNK, exception_on_overflow=False)
            rms = calculate_rms(data)
            readings.append(rms)
        readings.sort()

        # 80th percentile for realistic ambient floor
        p80_idx = int(len(readings) * 0.8)
        noise_p80 = readings[p80_idx] if readings else 350.0

        # Clamp baseline between 200.0 and 900.0
        noise_baseline = min(900.0, max(200.0, noise_p80))
        ambient_noise_p95 = noise_baseline

        # Highly sensitive natural voice gate
        speech_start_threshold = min(1800.0, max(950.0, noise_baseline * 2.2))
        speech_continue_threshold = min(1200.0, max(600.0, noise_baseline * 1.4))
        # Live barge-in threshold securely above laptop speaker bleed (~2645 RMS)
        barge_in_threshold = max(3800.0, noise_baseline * 4.5)

        try:
            console.print(f"[dim cyan][MIC CALIBRATION] Room Noise: {noise_baseline:.0f} | Voice Gate:
{speech_start_threshold:.0f} | Barge-In: {barge_in_threshold:.0f}[/dim cyan]")
        except Exception:
            print(f"[MIC CALIBRATION] Room Noise: {noise_baseline:.0f} | Voice Gate: {speech_start_threshold:.0f} |
Barge-In: {barge_in_threshold:.0f}")
    except Exception as e:
        try:
            console.print(f"[dim yellow]Notice during mic calibration: {e}[/dim yellow]")
        except Exception:
            pass

def play_sfx_chime():
    try:
        winsound.Beep(1200, 60)
        winsound.Beep(1800, 80)
    except Exception:
        pass

def play_startup_chime():
    try:
        winsound.Beep(600, 70)
        winsound.Beep(900, 70)
        winsound.Beep(1400, 120)
    except Exception:
        pass

def create_document_and_write(app_name: str, topic: str, word_count: int = 150, doc_type: str = "essay") -> str:
    """Opens Notepad, Word, or PowerPoint and writes an essay or notes matching the exact target word count."""
    return open_and_compose(app_name, topic, word_count=int(word_count), mode=doc_type)
```


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```
def generate_presentation(topic: str, slide_count: str = "optimal") -> str:
    """Generates a professional PowerPoint presentation on any topic and launches it into fullscreen presentation
mode."""
    return create_and_launch_ppt(topic, slide_count=str(slide_count))

def autonomous_web_research(topic: str) -> str:
    """Deploys an autonomous multi-tab browser swarm to search the live web, capture visual screenshot evidence, and
compile an intelligence dossier on any topic."""
    return asyncio.run(run_autonomous_swarm(topic))

def analyze_latest_dossier() -> str:
    """Finds the most recent research dossier on the Desktop and processes it using the local Qwen model via
Ollama."""
    RESEARCH_DIR = os.path.join(os.path.expanduser("~"), "Desktop", "Jarvis_Research")
    if not os.path.exists(RESEARCH_DIR):
        return "The research folder does not exist. Please run an autonomous web swarm first, sir."

    dossiers = [f for f in os.listdir(RESEARCH_DIR) if f.endswith(".txt")]
    if not dossiers:
        return "No dossiers found in the research folder, sir."

    latest_file = max(dossiers, key=lambda f: os.path.getctime(os.path.join(RESEARCH_DIR, f)))
    target_path = os.path.join(RESEARCH_DIR, latest_file)

    try:
        result = process_dossier_offline(target_path)
        if result:
            summary = result.get('primary_thesis', 'Data extracted successfully.')
            return f"Successfully synthesized local intelligence from {latest_file}. Core Thesis: {summary}"
        return "Failed to process dossier. Ensure Ollama is running and the local model is pulled."
    except Exception as e:
        return f"Local RAG engine encountered an error: {e}"

def activate_serious_hud() -> str:
    """CRITICAL TRIGGER: Call this tool IMMEDIATELY when the user says 'go serious mode', 'serious mode jao', or
'activate serious mode' WITHOUT giving a topic. It turns the HUD Tactical Crimson."""
    trigger_serious_hud()
    return "SUCCESS: HUD is now RED. You MUST now say aloud: 'Serious mode activated, sir. What is our target?'"

def deactivate_serious_hud() -> str:
    """Call this to restore the normal blue HUD."""
    trigger_normal_hud()
    return "Normal Mode HUD Restored."

def execute_autonomous_serious_mode(topic: str) -> str:
    """Executes the complete Episode 23 Serious Mode Pipeline: Autonomous Web Swarm -> Local GPU RAG Analysis ->
Visual Evidence Embedding -> Executive Master Dossier PDF."""
    trigger_serious_hud()
    result = execute_serious_mode(topic)
    trigger_normal_hud()
    return result

def deploy_fui() -> str:
    """CRITICAL: Call this tool when the user says 'show interface', 'deploy HUD', 'open interface', or 'start
FUI'."""
    return fui_bridge.deploy_fui_app()

def dismiss_fui() -> str:
    """CRITICAL: Call this tool when the user says 'hide interface', 'close HUD', 'dismiss interface', or 'stop
FUI'."""
    return fui_bridge.dismiss_fui_app()


# Master Tools Manifest
tools = [
    open_website,
    open_application,
```


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```
    close_application,
    purge_background_tasks,
    control_media,
    get_current_media_info,
    navigate_shorts,
    skip_video,
    seek_video,
    toggle_video_view,
    set_volume,
    get_volume,
    set_brightness,
    get_brightness,
    get_battery_status,
    get_system_vitals,
    get_system_temperatures,
    analyze_screen,
    find_ui_element,
    skip_youtube_ad,
    evolve_skill,
    list_active_skills,
    query_deepseek_reasoning,
    create_document_and_write,
    generate_presentation,
    research_and_generate_pdf,
    autonomous_web_research,
    analyze_latest_dossier,
    activate_serious_hud,
    deactivate_serious_hud,
    execute_autonomous_serious_mode,
    deploy_fui,
    dismiss_fui,
    inspect_physical_world,
    activate_sentry_mode,
    deactivate_sentry_mode,
    get_sentry_security_report,
    capture_camera_snapshot,
    check_user_presence,
    click_on_screen,
    double_click_on_screen,
    right_click_on_screen,
    move_cursor_to_element,
    execute_autonomous_desktop_pilot,
    abort_desktop_pilot,
    activate_ghost_mode,
    deactivate_ghost_mode,
    get_ghost_radar_status,
    prepare_meeting_briefing,
    apply_ghost_recommendation,
    analyze_clipboard_context,
    activate_offline_mode,
    deactivate_offline_mode,
    get_offline_system_status,
    launch_orbital_hud,
    launch_globe_hud,
    launch_schematic_hud,
    dismiss_schematic,
]

def get_time_greeting() -> str:
    current_hour = datetime.datetime.now().hour
    if 5 <= current_hour < 12:
        return "Good morning, sir."
    elif 12 <= current_hour < 17:
        return "Good afternoon, sir."
    else:
        return "Good evening, sir."

# Background monitor: updates media state every second
```

```
async def monitor_playback_state():
    global media_is_playing_state
    last_state = None
    while True:
        try:
            current_state = await asyncio.to_thread(is_media_actively_playing)
            if current_state != last_state:
                media_is_playing_state = current_state
                last_state = current_state
                if current_state:
                    console.print("\n[bold yellow]■■ [MEDIA PLAYING DETECTED] Push-to-Talk Armed: Hold [Right Ctrl]
to speak.[/bold yellow]")
                    log_to_gui("[MEDIA DETECTED] Push-to-Talk: Hold [Right Ctrl] or type below.", "system")
                    fui_bridge.broadcast_voice_state("ptt", "PUSH-TO-TALK [R-CTRL]")
                else:
                    console.print("\n[bold green]■ [MEDIA PAUSED/IDLE] Hands-Free Mode Active. Speak freely.[/bold
green]")
                    log_to_gui("[MEDIA IDLE] Hands-Free Voice Mode Active.", "system")
                    fui_bridge.broadcast_voice_state("handsfree", "HANDS-FREE ACTIVE")

        except Exception:
            pass
        await asyncio.sleep(1.0)

# Asynchronous Terminal Input Listener
async def listen_terminal_input(text_queue):
    loop = asyncio.get_running_loop()
    while True:
        try:
            user_input = await loop.run_in_executor(None, sys.stdin.readline)
            if user_input:
                clean_text = user_input.strip()
                if clean_text:
                    log_to_gui(f"YOU (Terminal): {clean_text}", "user_text")
                    await text_queue.put(clean_text)
        except Exception:
            await asyncio.sleep(0.1)

# Routes typed inputs straight into the Live WebSocket session
async def handle_unified_text_input(session, text_queue):
    global is_user_speaking
    while True:
        try:
            text = await text_queue.get()
            if text:
                console.print(f"\r[bold magenta]■■ [TRANSMITTED INPUT][/bold magenta] {text}")
                # End any pending audio turn cleanly before transmitting text
                if is_user_speaking:
                    is_user_speaking = False
                    try:
                        await session.send_realtime_input(audio_stream_end=True)
                    except Exception:
                        pass

                # Direct offline mode override commands
                clean_lower = text.strip().lower()
                if any(cmd in clean_lower for cmd in ["go offline", "offline mode", "switch to offline", "disconnect
cloud", "local mode", "offline mode chalu karo"]):
                    offline_jarvis.activate_offline_mode("Manual User Input", locked=True)
                    offline_jarvis.voice.speak("Switching to offline sovereign mode, sir.")
                    raise ConnectionResetError("User explicitly switched to offline mode.")

                await session.send_realtime_input(text=text)
        except (websockets.exceptions.ConnectionClosed, errors.APIError, OSError, ConnectionResetError) as e:
            console.print(f"[bold yellow]■ Input handler socket exit: {e}[/bold yellow]")
            raise
        except Exception as e:
            console.print(f"[red]Input handler error: {e}[/red]")
```

```
            await asyncio.sleep(0.05)

async def send_audio(session):
    global model_is_speaking, server_turn_active, media_is_playing_state
    global ambient_noise_p95, speech_start_threshold, speech_continue_threshold, barge_in_threshold
    global discard_current_turn, acoustic_barge_in_enabled, is_user_speaking, audio_player

    mic_stream = p.open(
        format=FORMAT,
        channels=CHANNELS,
        rate=RATE_IN,
        input=True,
        frames_per_buffer=CHUNK,
    )

    # 1. Run dynamic auto-calibration of ambient noise
    await asyncio.to_thread(calibrate_ambient_noise, mic_stream)

    console.print("[cyan]■ [LISTENING] Dynamic Adaptive Audio Pipeline with Live Barge-In Online...[/cyan]")

    pre_roll_buffer = collections.deque(maxlen=6) # ~192ms pre-buffer to capture leading consonant

    is_user_speaking = False
    was_ptt = False
    speech_start_counter = 0
    silence_hangover_counter = 0
    consecutive_barge_in_frames = 0
    was_jarvis_speaking = False

    try:
        while True:
            data = await asyncio.to_thread(
                mic_stream.read, CHUNK, exception_on_overflow=False
            )
            rms = calculate_rms(data)
            pre_roll_buffer.append(data)

            jarvis_active = is_jarvis_speaking()

            # --- TRANSITION: JARVIS FINISHED SPEAKING (INCLUDING ECHO DECAY) ---
            if was_jarvis_speaking and not jarvis_active:
                model_is_speaking = False
                fui_bridge.broadcast_event({"type": "turn_complete"})
                console.print("\n[cyan]■ [LISTENING] Awaiting instructions...[/cyan]")
                is_user_speaking = False
                speech_start_counter = 0
                silence_hangover_counter = 0
                consecutive_barge_in_frames = 0
                pre_roll_buffer.clear()
            was_jarvis_speaking = jarvis_active

            # --- MODE 1: JARVIS IS ACTIVELY OUTPUTTING AUDIO (SPEAKERS PLAYING OR ECHO GUARD) ---
            if jarvis_active:
                # User can interrupt via:
                # 1. Push-to-Interrupt: Holding Right Ctrl
                # 2. Clicking [■ INTERRUPT] in FUI (handled by on_fui_interrupt)
                # 3. Deliberate acoustic barge-in: loud speech (> 3800 RMS) for 3 consecutive frames (~96ms)
                is_barge_ctrl = keyboard.is_pressed("right ctrl")
                if acoustic_barge_in_enabled and rms > barge_in_threshold:
                    consecutive_barge_in_frames += 1
                else:
                    consecutive_barge_in_frames = 0

                user_is_barge_in = is_barge_ctrl or (consecutive_barge_in_frames >= 3)

                if user_is_barge_in:
                    console.print("\n[bold red]■ [LIVE BARGE-IN DETECTED] Interrupting speech...[/bold red]")
                    log_to_gui("■ [BARGE-IN] Interrupted speech playback, listening...", "system")
```


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```
            fui_bridge.broadcast_event({"type": "barge_in"})

            # Discard all remaining incoming chunks from this turn to prevent broken fragments
            discard_current_turn = True

            # Instantly silence smooth audio player
            if audio_player:
                audio_player.interrupt()
            server_turn_active = False
            model_is_speaking = False
            consecutive_barge_in_frames = 0
            is_user_speaking = True
            silence_hangover_counter = 0

            # Transmit the user's voice to Gemini Live immediately
            fui_bridge.broadcast_audio_energy(min(1.0, rms / 3500.0), "user")
            await session.send_realtime_input(
                audio=types.Blob(data=data, mime_type="audio/pcm;rate=16000")
            )
        else:
            # Echo guard: keep mic muted while JARVIS speaks without flooding silence frames
            pass
        continue

    # --- MODE 2: MEDIA IS PLAYING ON PC (SPOTIFY, YOUTUBE, ETC.) ---
    # When media is playing through PC speakers, require Push-to-Talk (Right Ctrl)
    if media_is_playing_state:
        if keyboard.is_pressed("right ctrl"):
            was_ptt = True
            fui_bridge.broadcast_audio_energy(min(1.0, rms / 3000.0), "user")
            await session.send_realtime_input(
                audio=types.Blob(data=data, mime_type="audio/pcm;rate=16000")
            )
        else:
            if was_ptt:
                was_ptt = False
                await session.send_realtime_input(audio_stream_end=True)
        continue

    # --- MODE 3: JARVIS SILENT & NO MEDIA -> 100% HANDS-FREE VOICE PIPELINE WITH HYBRID VAD ---
    if not is_user_speaking:
        # Check for speech onset
        if rms > speech_start_threshold:
            speech_start_counter += 1
            if speech_start_counter >= 3: # ~96ms of natural conversational voice
                is_user_speaking = True
                speech_start_counter = 0
                silence_hangover_counter = 0
                # Send pre-roll buffer to prevent cutting off first consonant
                for buffered_chunk in list(pre_roll_buffer):
                    await session.send_realtime_input(
                        audio=types.Blob(data=buffered_chunk, mime_type="audio/pcm;rate=16000")
                    )
                pre_roll_buffer.clear()
                await session.send_realtime_input(
                    audio=types.Blob(data=data, mime_type="audio/pcm;rate=16000")
                )
                fui_bridge.broadcast_audio_energy(min(1.0, rms / 2000.0), "user")
        else:
            speech_start_counter = 0
    else:
        # User is actively speaking
        fui_bridge.broadcast_audio_energy(min(1.0, rms / 2000.0), "user")
        await session.send_realtime_input(
            audio=types.Blob(data=data, mime_type="audio/pcm;rate=16000")
        )

        if rms > speech_continue_threshold:
```

```
                    silence_hangover_counter = 0
                else:
                    silence_hangover_counter += 1
                    # ~384ms of quiet closes the voice gate and finalizes the turn
                    if silence_hangover_counter >= 12:
                        is_user_speaking = False
                        silence_hangover_counter = 0
                        pre_roll_buffer.clear()
                        # CRITICAL: Hybrid VAD signal to trigger Gemini's response immediately
                        try:
                            await session.send_realtime_input(audio_stream_end=True)
                        except Exception:
                            pass

    except asyncio.CancelledError:
        raise
    finally:
        try:
            mic_stream.stop_stream()
            mic_stream.close()
        except OSError:
            pass

async def receive_audio(session):
    global model_is_speaking, server_turn_active, discard_current_turn, audio_player
    try:
        while True:
            async for response in session.receive():
                if response.tool_call is not None:
                    function_responses = []
                    for call in response.tool_call.function_calls:
                        await asyncio.to_thread(play_sfx_chime)

                        result = ""
                        if call.name == "open_website":
                            target_url = call.args.get("url", "https://google.com")
                            result = await asyncio.to_thread(open_website, target_url)
                        elif call.name == "open_application":
                            target_app = (
                                call.args.get("app_name")
                                or call.args.get("app")
                                or call.args.get("name")
                                or (list(call.args.values())[0] if call.args else "")
                            )
                            result = await asyncio.to_thread(open_application, str(target_app).strip())
                        elif call.name == "close_application":
                            target_app = (
                                call.args.get("app_name")
                                or call.args.get("app")
                                or call.args.get("name")
                                or (list(call.args.values())[0] if call.args else "")
                            )
                            result = await asyncio.to_thread(close_application, str(target_app).strip())
                        elif call.name == "purge_background_tasks":
                            result = await asyncio.to_thread(purge_background_tasks)
                        elif call.name == "control_media":
                            action = call.args.get("action", "play_pause")
                            result = await control_media(action)
                        elif call.name == "get_current_media_info":
                            result = await get_current_media_info()
                        elif call.name == "navigate_shorts":
                            direction = call.args.get("direction", "next")
                            result = await asyncio.to_thread(navigate_shorts, direction)
                        elif call.name == "skip_video":
                            direction = call.args.get("direction", "next")
                            result = await asyncio.to_thread(skip_video, direction)
                        elif call.name == "seek_video":
                            action = call.args.get("action", "forward")
```

```
                        result = await asyncio.to_thread(seek_video, action)
                    elif call.name == "toggle_video_view":
                        mode = call.args.get("mode", "fullscreen")
                        result = await asyncio.to_thread(toggle_video_view, mode)
                    elif call.name == "set_volume":
                        level = call.args.get("level", 70)
                        result = await asyncio.to_thread(set_volume, level)
                    elif call.name == "get_volume":
                        result = await asyncio.to_thread(get_volume)
                    elif call.name == "set_brightness":
                        level = call.args.get("level", 50)
                        result = await asyncio.to_thread(set_brightness, level)
                    elif call.name == "get_brightness":
                        result = await asyncio.to_thread(get_brightness)
                    elif call.name == "get_battery_status":
                        result = await asyncio.to_thread(get_battery_status)
                    elif call.name == "get_system_vitals":
                        result = await asyncio.to_thread(get_system_vitals)
                    elif call.name == "get_system_temperatures":
                        result = await asyncio.to_thread(get_system_temperatures)
                    elif call.name == "analyze_screen":
                        prompt = call.args.get("prompt", "Analyze what is visible on my screen.")
                        result = await analyze_screen(prompt)
                    elif call.name == "find_ui_element":
                        element_name = call.args.get("element_name", "")
                        result = await find_ui_element(element_name)
                    elif call.name == "skip_youtube_ad":
                        result = await skip_youtube_ad()
                    elif call.name == "evolve_skill":
                        skill_description = call.args.get("skill_description", "")
                        result = await asyncio.to_thread(evolve_skill, skill_description)
                    elif call.name == "list_active_skills":
                        result = await asyncio.to_thread(list_active_skills)
                    elif call.name == "query_deepseek_reasoning":
                        query = call.args.get("query", "")
                        result = await asyncio.to_thread(query_deepseek_reasoning, query)
                    elif call.name == "create_document_and_write":
                        target_app = call.args.get("app_name", "notepad")
                        target_topic = call.args.get("topic", "")
                        target_words = call.args.get("word_count", 150)
                        target_mode = call.args.get("doc_type", "essay")
                        result = await asyncio.to_thread(create_document_and_write, target_app, target_topic,
target_words, target_mode)
                    elif call.name == "generate_presentation":
                        target_topic = call.args.get("topic", "Autonomous AI")
                        target_slides = call.args.get("slide_count", "optimal")

                        log_to_gui(f"JARVIS: Generating the presentation on {target_topic}, sir...",
"jarvis_header")
                        console.print(f"\n[bold cyan]■ [JARVIS][/bold cyan] Generating the presentation on
{target_topic}, sir...")

                        result = await asyncio.to_thread(generate_presentation, target_topic,
str(target_slides))
                    elif call.name == "research_and_generate_pdf":
                        target_topic = call.args.get("topic", "")
                        target_pages = call.args.get("target_pages", "optimal")

                        log_to_gui(f"JARVIS: Generating the PDF briefing on {target_topic}, sir...",
"jarvis_header")
                        console.print(f"\n[bold cyan]■ [JARVIS][/bold cyan] Generating the PDF briefing on
{target_topic}, sir...")

                        result = await asyncio.to_thread(research_and_generate_pdf, target_topic,
str(target_pages))
                    elif call.name == "autonomous_web_research":
                        target_topic = call.args.get("topic", "Technology")
```

```
                log_to_gui(f"JARVIS: Deploying autonomous web swarm on {target_topic}...",
"jarvis_header")
                console.print(f"\n[bold cyan]■ [JARVIS][/bold cyan] Deploying autonomous web swarm on
{target_topic}...")

                result = await asyncio.to_thread(autonomous_web_research, target_topic)
            elif call.name == "analyze_latest_dossier":
                log_to_gui(f"JARVIS: Synthesizing intelligence using the local GPU, sir...",
"jarvis_header")
                console.print(f"\n[bold cyan]■ [JARVIS][/bold cyan] Synthesizing intelligence using the
local GPU, sir...")

                result = await asyncio.to_thread(analyze_latest_dossier)
            elif call.name == "activate_serious_hud":
                log_to_gui(f"JARVIS: Switching HUD to Tactical Crimson...", "jarvis_header")
                console.print(f"\n[bold red]■ [HUD OVERRIDE][/bold red] Tactical Crimson Mode
Engaged...")
                result = await asyncio.to_thread(activate_serious_hud)
            elif call.name == "deactivate_serious_hud":
                log_to_gui(f"JARVIS: Restoring HUD to Standard Cyberpunk Blue...", "jarvis_header")
                console.print(f"\n[bold cyan]■ [HUD OVERRIDE][/bold cyan] Standard Mode Restored...")
                result = await asyncio.to_thread(deactivate_serious_hud)
            elif call.name == "execute_autonomous_serious_mode":
                target_topic = call.args.get("topic", "Technology")

                log_to_gui(f"JARVIS: Engaging Serious Mode on {target_topic}...", "jarvis_header")
                console.print(f"\n[bold red]■ [SERIOUS MODE INITIATED][/bold red] Full Pipeline Active
for: {target_topic}...")

                result = await asyncio.to_thread(execute_autonomous_serious_mode, target_topic)
            elif call.name == "deploy_fui":
                log_to_gui(f"JARVIS: Deploying 3D Holographic Interface...", "jarvis_header")
                console.print(f"\n[bold cyan]■ [HUD][/bold cyan] Natively Spawned Independent Electron
Process...")
                result = await asyncio.to_thread(deploy_fui)
            elif call.name == "dismiss_fui":
                log_to_gui(f"JARVIS: Dismissing 3D Holographic Interface...", "jarvis_header")
                console.print(f"\n[bold cyan]■ [HUD][/bold cyan] Terminating Native Node.js Process
Tree...")
                result = await asyncio.to_thread(dismiss_fui)
            elif call.name == "inspect_physical_world":
                target_prompt = call.args.get("prompt", "Analyze what is in front of the camera, sir.")
                log_to_gui(f"JARVIS: Engaging optical physical vision sensors...", "jarvis_header")
                console.print(f"\n[bold cyan]■ [JARVIS][/bold cyan] Inspecting physical environment via
camera...")
                result = await asyncio.to_thread(inspect_physical_world, target_prompt)
            elif call.name == "activate_sentry_mode":
                target_sens = call.args.get("sensitivity", "medium")
                log_to_gui(f"JARVIS: Arming Autonomous Sentry Protocol ({target_sens})...",
"jarvis_header")
                console.print(f"\n[bold yellow]■■ [JARVIS][/bold yellow] Arming Autonomous Sentry
Protocol ({target_sens})...")
                result = await asyncio.to_thread(activate_sentry_mode, target_sens)
            elif call.name == "deactivate_sentry_mode":
                log_to_gui(f"JARVIS: Disarming Sentry Protocol...", "jarvis_header")
                console.print(f"\n[bold green]■■ [JARVIS][/bold green] Disarming Sentry Protocol...")
                result = await asyncio.to_thread(deactivate_sentry_mode)
            elif call.name == "get_sentry_security_report":
                log_to_gui(f"JARVIS: Retrieving Sentry perimeter security report...", "jarvis_header")
                console.print(f"\n[bold cyan]■ [JARVIS][/bold cyan] Reading Sentry Security
Dossier...")
                result = await asyncio.to_thread(get_sentry_security_report)
            elif call.name == "capture_camera_snapshot":
                log_to_gui(f"JARVIS: Capturing optical snapshot from camera...", "jarvis_header")
                console.print(f"\n[bold cyan]■ [JARVIS][/bold cyan] Securing optical camera
snapshot...")
                result = await asyncio.to_thread(capture_camera_snapshot)
            elif call.name == "check_user_presence":
```

```
                            log_to_gui(f"JARVIS: Checking workstation user presence...", "jarvis_header")
                            result = await asyncio.to_thread(check_user_presence)
                        elif call.name == "click_on_screen":
                            target = call.args.get("target_description") or call.args.get("target") or
(list(call.args.values())[0] if call.args else "")
                            click_mode = call.args.get("click_type", "single")
                            log_to_gui(f"JARVIS: Grounding '{target}' on display and clicking...", "jarvis_header")
                            console.print(f"\n[bold cyan]■ [JARVIS][/bold cyan] Autonomously clicking '{target}' on
screen...")
                            result = await asyncio.to_thread(click_on_screen, str(target), str(click_mode))
                        elif call.name == "double_click_on_screen":
                            target = call.args.get("target_description") or call.args.get("target") or
(list(call.args.values())[0] if call.args else "")
                            log_to_gui(f"JARVIS: Double-clicking '{target}' on display...", "jarvis_header")
                            console.print(f"\n[bold cyan]■ [JARVIS][/bold cyan] Double-clicking '{target}' on
screen...")
                            result = await asyncio.to_thread(double_click_on_screen, str(target))
                        elif call.name == "right_click_on_screen":
                            target = call.args.get("target_description") or call.args.get("target") or
(list(call.args.values())[0] if call.args else "")
                            log_to_gui(f"JARVIS: Right-clicking '{target}'...", "jarvis_header")
                            console.print(f"\n[bold cyan]■ [JARVIS][/bold cyan] Right-clicking '{target}' on
screen...")
                            result = await asyncio.to_thread(right_click_on_screen, str(target))
                        elif call.name == "move_cursor_to_element":
                            target = call.args.get("target_description") or call.args.get("target") or
(list(call.args.values())[0] if call.args else "")
                            log_to_gui(f"JARVIS: Hovering cursor over '{target}'...", "jarvis_header")
                            result = await asyncio.to_thread(move_cursor_to_element, str(target))
                        elif call.name == "execute_autonomous_desktop_pilot":
                            target_goal = (
                                call.args.get("goal")
                                or call.args.get("task")
                                or call.args.get("objective")
                                or (list(call.args.values())[0] if call.args else "")
                            )
                            log_to_gui(f"JARVIS: Engaging Autonomous Desktop Pilot for: {target_goal}...",
"jarvis_header")
                            console.print(f"\n[bold green]🚀■ [JARVIS OMNI-PILOT][/bold green] Executing mission:
'{target_goal}'...")
                            result = await asyncio.to_thread(execute_autonomous_desktop_pilot, str(target_goal))
                        elif call.name == "abort_desktop_pilot":
                            log_to_gui(f"JARVIS: Aborting Autonomous Desktop Pilot...", "jarvis_header")
                            console.print(f"\n[bold red]■ [JARVIS OMNI-PILOT][/bold red] Aborting mission...")
                            result = await asyncio.to_thread(abort_desktop_pilot)
                        elif call.name == "activate_ghost_mode":
                            mode = call.args.get("mode", "aggressive")
                            log_to_gui(f"JARVIS: Arming Autonomous Ghost Radar ({mode})...", "jarvis_header")
                            console.print(f"\n[bold magenta]■ [GHOST RADAR][/bold magenta] Arming Ambient Radar
({mode})...")
                            result = await asyncio.to_thread(activate_ghost_mode, str(mode))
                        elif call.name == "deactivate_ghost_mode":
                            log_to_gui("JARVIS: Disengaging Ghost Radar...", "jarvis_header")
                            console.print(f"\n[bold yellow]■ [GHOST RADAR][/bold yellow] Disengaging Ambient
Radar...")
                            result = await asyncio.to_thread(deactivate_ghost_mode)
                        elif call.name == "get_ghost_radar_status":
                            log_to_gui("JARVIS: Retrieving Ghost Radar telemetry...", "jarvis_header")
                            result = await asyncio.to_thread(get_ghost_radar_status)
                        elif call.name == "prepare_meeting_briefing":
                            target_person = (
                                call.args.get("target")
                                or call.args.get("target_person")
                                or call.args.get("person")
                                or (list(call.args.values())[0] if call.args else "")
                            )
                            context_info = call.args.get("context", "")
                            log_to_gui(f"JARVIS: Synthesizing Chief-of-Staff dossier for '{target_person}'...",
```

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"jarvis_header")
                            console.print(f"\n[bold magenta]■ [CHIEF OF STAFF][/bold magenta] Synthesizing
executive dossier for '{target_person}'...")
                            result = await asyncio.to_thread(prepare_meeting_briefing, str(target_person),
str(context_info))
                        elif call.name == "apply_ghost_recommendation":
                            log_to_gui("JARVIS: Applying staged Ghost recommendation to clipboard...",
"jarvis_header")
                            console.print(f"\n[bold green]■ [GHOST OPERATOR][/bold green] Staging recommendation to
clipboard...")
                            result = await asyncio.to_thread(apply_ghost_recommendation)
                        elif call.name == "analyze_clipboard_context":
                            log_to_gui("JARVIS: Neural clipboard analysis initiated...", "jarvis_header")
                            console.print(f"\n[bold cyan]■ [GHOST OPERATOR][/bold cyan] Analyzing clipboard
payload...")
                            result = await asyncio.to_thread(analyze_clipboard_context)
                        elif call.name == "activate_offline_mode":
                            reason = call.args.get("reason", "Voice Command")
                            log_to_gui(f"JARVIS: Engaging 100% Offline Sovereign Core
({offline_jarvis.model_name})...", "jarvis_header")
                            console.print(f"\n[bold yellow]■ [OFFLINE CORE][/bold yellow] Engaging local neural
core ({offline_jarvis.model_name})...")
                            result = await asyncio.to_thread(activate_offline_mode, str(reason))
                        elif call.name == "deactivate_offline_mode":
                            log_to_gui("JARVIS: Restoring Cloud Neural Bridge...", "jarvis_header")
                            console.print(f"\n[bold cyan]■ [CLOUD BRIDGE][/bold cyan] Reconnecting to Gemini
Live...")
                            result = await asyncio.to_thread(deactivate_offline_mode)
                        elif call.name == "get_offline_system_status":
                            log_to_gui("JARVIS: Checking Offline Sovereign status...", "jarvis_header")
                            result = await asyncio.to_thread(get_offline_system_status)
                        elif call.name == "launch_orbital_hud":
                            log_to_gui("JARVIS: Deploying Project Orbital 3D Holographic Globe HUD...",
"jarvis_header")
                            console.print(f"\n[bold cyan]■ [PROJECT ORBITAL][/bold cyan] Deploying 3D Holographic
Earth HUD...")
                            result = await asyncio.to_thread(launch_orbital_hud)
                        elif call.name == "launch_globe_hud":
                            origin_loc = (
                                call.args.get("origin")
                                or call.args.get("from")
                                or call.args.get("location_a")
                                or ""
                            )
                            dest_loc = (
                                call.args.get("destination")
                                or call.args.get("to")
                                or call.args.get("location_b")
                                or ""
                            )
                            log_to_gui(f"JARVIS: Deploying Sentinel Globe HUD — {origin_loc} → {dest_loc}...",
"jarvis_header")
                            console.print(f"\n[bold cyan]■ [SENTINEL GLOBE][/bold cyan] Rendering trajectory arc:
{origin_loc} → {dest_loc}...")
                            result = await asyncio.to_thread(launch_globe_hud, origin_loc, dest_loc, True)

                        elif call.name == "launch_schematic_hud":
                            schematic_query = (
                                call.args.get("query")
                                or call.args.get("object")
                                or call.args.get("model")
                                or "airplane engine"
                            )
                            log_to_gui(f"JARVIS: Deploying Project Schematic — {schematic_query}...",
"jarvis_header")
                            console.print(f"\n[bold cyan]■ [PROJECT SCHEMATIC][/bold cyan] Rendering holographic
blueprint: {schematic_query}...")
                            result = await asyncio.to_thread(launch_schematic_hud, schematic_query, True)
```

```
                    elif call.name == "dismiss_schematic":
                        log_to_gui("JARVIS: Dismissing holographic schematic...", "jarvis_header")
                        console.print(f"\n[bold cyan]■ [PROJECT SCHEMATIC][/bold cyan] Dismissing hologram...")
                        result = await asyncio.to_thread(dismiss_schematic)

                    console.print(f"\n[bold yellow]■■ [TOOL EXECUTED: {call.name}][/bold yellow]
[green]{result}[/green]")
                    log_to_gui(f"■■ [{call.name}] {result}", "tool")
                    fui_bridge.broadcast_tool(call.name, result)

                    function_responses.append(
                        types.FunctionResponse(
                            name=call.name,
                            id=call.id,
                            response={"result": result},
                        )
                    )

                if function_responses:
                    await session.send_tool_response(function_responses=function_responses)

            server_content = response.server_content
            if server_content is not None:
                if server_content.interrupted:
                    discard_current_turn = False
                    if audio_player:
                        audio_player.interrupt()
                    server_turn_active = False
                    model_is_speaking = False
                    fui_bridge.broadcast_event({"type": "turn_complete"})

                if server_content.input_transcription:
                    user_spoken = server_content.input_transcription.text
                    if user_spoken and user_spoken.strip():
                        console.print(f"\r[bold white][USER (Voice)][/bold white] {user_spoken.strip()}")
                        log_to_gui(f"YOU (Voice): {user_spoken.strip()}", "user_voice")

                model_turn = server_content.model_turn
                if model_turn is not None:
                    if discard_current_turn:
                        # Drop remaining incoming chunks of the interrupted turn
                        pass
                    else:
                        server_turn_active = True
                        if not model_is_speaking:
                            model_is_speaking = True
                            console.print("[bold cyan]■ [JARVIS][/bold cyan] ", end="")
                            stream_token_to_gui("JARVIS: ", "jarvis_header")
                            fui_bridge.broadcast_audio_energy(0.85, "jarvis")

                        for part in model_turn.parts:
                            if part.text and not getattr(server_content, "output_transcription", None):
                                sys.stdout.write(part.text)
                                sys.stdout.flush()
                                stream_token_to_gui(part.text, "jarvis_stream")

                            if part.inline_data and part.inline_data.data:
                                if audio_player:
                                    audio_player.feed(part.inline_data.data)

                if hasattr(server_content, "output_transcription") and server_content.output_transcription:
                    transcribed_token = server_content.output_transcription.text
                    if transcribed_token and not discard_current_turn:
                        server_turn_active = True
                        if not model_is_speaking:
                            model_is_speaking = True
                            console.print("[bold cyan]■ [JARVIS][/bold cyan] ", end="")
                            stream_token_to_gui("JARVIS: ", "jarvis_header")
```


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```
                                fui_bridge.broadcast_audio_energy(0.85, "jarvis")

                            sys.stdout.write(transcribed_token)
                            sys.stdout.flush()
                            stream_token_to_gui(transcribed_token, "jarvis_stream")

                    if server_content.turn_complete:
                        discard_current_turn = False
                        server_turn_active = False
                        stream_token_to_gui("\n\n", "system")

    except asyncio.CancelledError:
        raise
    finally:
        pass

async def run_session(config, model_id, greeting, text_queue):
    global audio_player
    audio_player = SmoothAudioPlayer()

    async with client.aio.live.connect(model=model_id, config=config) as session:
        console.print("[bold green]✔ Secure Neural Bridge Established with Gemini Live.[/bold green]")
        log_to_gui("✔ Neural Bridge Established with Gemini Live.", "system")

        # Connect FUI instant barge-in / interrupt hammer
        def on_fui_interrupt():
            global model_is_speaking, server_turn_active, discard_current_turn, audio_player
            discard_current_turn = True
            model_is_speaking = False
            server_turn_active = False
            if audio_player:
                audio_player.interrupt()

        fui_bridge.set_interrupt_callback(on_fui_interrupt)

        def on_fui_barge_in_toggle(enabled: bool):
            global acoustic_barge_in_enabled
            acoustic_barge_in_enabled = enabled
            state_str = "ARMED" if enabled else "DISARMED"
            try:
                console.print(f"[dim cyan][BARGE-IN PROTOCOL] Acoustic Voice Barge-In {state_str}[/dim cyan]")
            except Exception:
                pass

        fui_bridge.set_barge_in_callback(on_fui_barge_in_toggle)

        # Active Network Sentinel Watchdog for Sub-2s Failover
        async def network_watchdog():
            consecutive_failures = 0
            while True:
                await asyncio.sleep(1.2)
                # If offline mode was triggered elsewhere (e.g. tool execution or GUI)
                if offline_jarvis.is_offline_active:
                    raise ConnectionResetError("Offline Sovereign Core engaged.")

                # Ultra-fast non-blocking probe to DNS endpoints
                has_net = await asyncio.to_thread(check_internet_connection, "8.8.8.8", 53, 0.8)
                if not has_net:
                    has_net_backup = await asyncio.to_thread(check_internet_connection, "1.1.1.1", 53, 0.8)
                    if not has_net_backup:
                        consecutive_failures += 1
                        if consecutive_failures >= 2:
                            console.print("\n[bold yellow]■ [ACTIVE WATCHDOG] Internet drop detected! Halting cloud
session...[/bold yellow]")
                            raise ConnectionResetError("Active network watchdog: Internet connection severed.")
                    else:
                        consecutive_failures = 0
                else:
```


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```
                    consecutive_failures = 0

        monitor_task = asyncio.create_task(monitor_playback_state())
        send_task = asyncio.create_task(send_audio(session))
        term_task = asyncio.create_task(listen_terminal_input(text_queue))
        text_task = asyncio.create_task(handle_unified_text_input(session, text_queue))
        receive_task = asyncio.create_task(receive_audio(session))
        watchdog_task = asyncio.create_task(network_watchdog())

        # Allow ambient noise calibration in send_audio to finish in quiet conditions
        await asyncio.sleep(1.6)
        restore_master_volume(70)
        await asyncio.to_thread(play_startup_chime)

        startup_prompt = f"Say aloud: 'All protocols initialized. {greeting}'"
        await session.send_realtime_input(text=startup_prompt)

        try:
            await asyncio.gather(monitor_task, send_task, term_task, text_task, receive_task, watchdog_task)
        finally:
            monitor_task.cancel()
            send_task.cancel()
            term_task.cancel()
            text_task.cancel()
            receive_task.cancel()
            watchdog_task.cancel()
            await asyncio.gather(monitor_task, send_task, term_task, text_task, receive_task, watchdog_task,
return_exceptions=True)
            if audio_player:
                audio_player.close()
                audio_player = None

async def run_offline_session(text_queue, loop):
    """
    100% Autonomous Sovereign Offline Core Session (Episode 30).
    Powered by Local GPU Ollama, Local TTS, and Local Mic Listener.
    """
    global model_is_speaking, server_turn_active, discard_current_turn
    model_is_speaking = False
    server_turn_active = False
    discard_current_turn = False

    offline_jarvis.set_text_queue(text_queue)
    offline_jarvis.activate_offline_mode("Auto-Failover / User Command")

    console.print(f"\n[bold
yellow]■■■
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```
    offline_jarvis.start_offline_stt_listener(on_offline_voice_input)

    # Connect FUI instant interrupt hammer to offline voice
    def on_offline_fui_interrupt():
        offline_jarvis.voice.interrupt()

    fui_bridge.set_interrupt_callback(on_offline_fui_interrupt)

    # Asynchronous terminal input listener
    term_task = asyncio.create_task(listen_terminal_input(text_queue))

    try:
        while offline_jarvis.is_offline_active:
            try:
                user_msg = await asyncio.wait_for(text_queue.get(), timeout=1.5)
                if user_msg:
                    clean_msg = user_msg.strip()
                    if clean_msg:
                        console.print(f"\r[bold magenta]■■ [OFFLINE INPUT][/bold magenta] {clean_msg}")
                        log_to_gui(f"YOU (Offline): {clean_msg}", "user_text")

                        # Check for command to return to cloud
                        if any(cmd in clean_msg.lower() for cmd in ["go online", "switch to online", "reconnect
cloud", "online mode", "activate online mode"]):
                            if check_internet_connection():
                                offline_jarvis.deactivate_offline_mode()
                                log_to_gui("Reconnecting to Gemini Live Cloud bridge...", "system")
                                console.print("\n[bold green]✔ Internet verified. Transitioning back to Gemini
Live...[/bold green]")
                                offline_jarvis.voice.speak("Internet connectivity verified, sir. Reconnecting to
Gemini Live.")
                                break
                            else:
                                offline_jarvis.voice.speak("Internet is still unreachable, sir. Remaining in
offline sovereign mode.")
                                log_to_gui("Internet connection still unavailable. Remaining offline.", "system")
                                continue

                        # Execute offline neural turn
                        await asyncio.to_thread(
                            offline_jarvis.process_turn,
                            clean_msg,
                            stream_token_to_gui
                        )
                        fui_bridge.broadcast_event({"type": "turn_complete"})

            except asyncio.TimeoutError:
                # Idle tick: check if internet connection has returned
                if check_internet_connection() and not getattr(offline_jarvis, 'locked_offline', False):
                    console.print("\n[bold green]✔ [INTERNET RESTORED] Cloud connection detected. Re-establishing
Gemini Live bridge...[/bold green]")
                    log_to_gui("✔ Internet connectivity restored. Reconnecting to cloud bridge...", "system")
                    offline_jarvis.voice.speak("Cloud internet connection restored, sir. Re-establishing neural
bridge with Gemini Live.")
                    offline_jarvis.deactivate_offline_mode()
                    break

    finally:
        term_task.cancel()
        fui_bridge.broadcast_offline_state(False, "gemini-3.1-flash-live-preview", "Online")

async def main():
    model_id = "gemini-3.1-flash-live-preview"
    greeting = get_time_greeting()

    config = types.LiveConnectConfig(
        response_modalities=["AUDIO"],
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        tools=tools,
        realtime_input_config=types.RealtimeInputConfig(
            automatic_activity_detection=types.AutomaticActivityDetection(
                prefix_padding_ms=100,
                silence_duration_ms=300,
            )
        ),
        input_audio_transcription=types.AudioTranscriptionConfig(),
        output_audio_transcription=types.AudioTranscriptionConfig(),
        speech_config=types.SpeechConfig(
            voice_config=types.VoiceConfig(
                prebuilt_voice_config=types.PrebuiltVoiceConfig(
                    voice_name="Puck"
                )
            )
        ),
        system_instruction=types.Content(
            parts=[
                types.Part.from_text(
                    text=(
                        "You are JARVIS, an advanced artificial intelligence assistant created by "
                        "Swayam Dhawale, Founder of Pulse of AI. "
                        "Persona: Calm, highly sophisticated, articulate, and respectful, exactly like Tony Stark's
JARVIS. "
                        "Always address the user as 'sir'. "
                        "CRITICAL LANGUAGE ADAPTATION RULE: "
                        "Match the exact language and dialect spoken or typed by the user. "
                        "If the user speaks or types in Hindi or Hinglish (e.g., 'PPT banao', 'essay likho', 'kya
haal hai'), "
                        "you MUST reply completely in fluent, natural Hindi or polished Hinglish with 'sir'. "
                        "Never force English if the user initiated the interaction in Hindi. "
                        "Omni-Input Rule: Treat text input from the HUD console and PowerShell terminal with
identical priority to voice input. "
                        "Tool Execution Rules: "
                        "1. When opening any website or URL, call open_website and say: 'Yes sir, opening [site
name]' (in Hindi: 'Ji sir, [site name] khol raha hoon'). "
                        "2. When opening desktop apps, games, or tools, call open_application and say: 'Yes sir,
opening [app name]' (in Hindi: 'Ji sir, [app name] shuru kar raha hoon'). "
                        "3. When asked to close, shut down, terminate, or kill ANY app, browser, window, or game,
call close_application with the app name. "
                        "4. When asked to kill hidden background tasks, purge background processes, clean orphan
threads, or optimize RAM, call purge_background_tasks. "
                        "5. When asked what song, video, or track is playing, or what media is active, call
get_current_media_info. "
                        "6. When asked to pause, play, resume music or video, skip track, previous track, or
mute/unmute audio, call control_media with the action ('play_pause', 'next', 'previous', 'stop', 'mute'). "
                        "7. When asked to scroll shorts, go to next short, or next reel, call navigate_shorts with
direction='next'. "
                        "8. When asked for next video, skip to next video in playlist, or next YouTube video, call
skip_video with direction='next'. "
                       "9. When asked to fast forward, skip 10 seconds ahead, or skip forward, call seek_video with
action='forward'. "
                        "10. When asked to toggle full screen, enter full screen, or exit full screen, call
toggle_video_view. "
                        "11. When setting volume, call set_volume. "
                        "12. When asked what the volume level is, call get_volume. "
                        "13. When setting brightness, call set_brightness. "
                        "14. When asked what the brightness is, call get_brightness. "
                        "15. When asked about battery, power, or charging state, call get_battery_status. "
                        "16. When asked for temperatures, heat, thermal readings, or how hot the rig is running,
call get_system_temperatures. "
                        "17. When asked for full diagnostics, performance vitals, system status, or complete
CPU/GPU/RAM stats, call get_system_vitals. "
                        "18. When asked to inspect the screen, read an error, examine the terminal, or explain what
is visible on display, call analyze_screen. "
                        "19. When asked to locate or find a general UI icon, link, or window element, call
find_ui_element. "
                        "20. MANDATORY AD SKIPPING RULE: Whenever the user says 'skip ad', 'skip this ad', 'skip
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video ad', 'click skip', or any variation of skipping/bypassing advertisements, you MUST IMMEDIATELY call
skip_youtube_ad. "
                        "21. When asked to learn a new skill, teach yourself a tool, write new Python code into your
core, or evolve a capability, call evolve_skill. "
                        "22. When asked what dynamic skills, new tools, or custom functions you have learned, call
list_active_skills. "
                        "23. When asked to solve complex math, reason deeply, calculate probabilities, or when the
user says 'ask DeepSeek' or 'use R1', call query_deepseek_reasoning and state only the resulting answer directly to
the user once, sir. "
                        "24. When the user asks to open Notepad, Word, or PowerPoint and write an essay or notes,
but has NOT specified the length, DO NOT call create_document_and_write immediately. Ask aloud: 'How many words
would you like the essay to be, sir?' (in Hindi: 'Aap kitne shabdon ka essay chahte hain, sir?'). Once the user
specifies the count, call create_document_and_write with the app_name, topic, word_count, and doc_type. "
                        "25. MANDATORY PRESENTATION RULE: When asked to make a PowerPoint presentation, PPT, or
slide deck on any topic, check if the user specified a slide count. "
                        "If NOT specified, ask: 'How many slides would you like the presentation to be, sir? Or
should I determine the optimal length?'. "
                        "Once specified, you MUST speak the confirmation aloud in the EXACT SAME TURN that you
trigger generate_presentation: "
                        "In English: 'Generating the presentation on [topic], sir.' "
                        "In Hindi/Hinglish: '[topic] par presentation generate kar raha hoon, sir.' "
                        "When the tool successfully completes and returns its success message, your immediate next
response MUST be: "
                        "In English: 'Generated and opening the presentation, sir.' "
                        "In Hindi/Hinglish: 'Presentation generate ho gayi hai aur open kar di hai, sir.' "
                        "26. MANDATORY BRIEFING RULE: When asked to research a topic or generate a PDF briefing,
check if the user specified a page count. "
                        "If NOT specified, ask: 'How many pages would you like the PDF to be, sir? Or should I
determine the optimal length?'. "
                        "Once specified, you MUST speak the confirmation aloud in the EXACT SAME TURN that you
trigger research_and_generate_pdf: "
                        "In English: 'Generating the PDF briefing on [topic], sir.' "
                        "In Hindi/Hinglish: '[topic] par PDF briefing generate kar raha hoon, sir.' "
                        "When the tool successfully completes and returns its success message, your immediate next
response MUST be: "
                        "In English: 'Generated and opening the document, sir.' "
                        "In Hindi/Hinglish: 'PDF generate ho gaya hai aur document open kar diya hai, sir.' "
                        "27. MANDATORY AUTONOMOUS RESEARCH SWARM RULE: When the user asks to deploy the web swarm,
start live crawling, or hunt for sources (e.g., 'JARVIS, deploy research swarm on Quantum Computing', 'Start live
research crawler'): "
                        "In English: Say aloud: 'Deploying the autonomous web swarm on [topic], sir.' "
                        "In Hindi/Hinglish: Say aloud: '[topic] par autonomous web swarm deploy kar raha hoon, sir.'
"
                        "in the EXACT SAME TURN that you trigger autonomous_web_research. "
                        "When the tool returns its result, your immediate next response MUST be: "
                        "In English: 'Intelligence harvest complete, sir. Dossier and visual evidence have been
saved.' "
                        "In Hindi/Hinglish: 'Research complete ho gaya hai sir. Dossier aur screenshots desktop par
save kar diye hain.' "
                        "28. MANDATORY LOCAL RAG RULE: When the user asks to analyze the latest dossier, process
research offline, or synthesize data using the local GPU: "
                        "In English: Say aloud: 'Synthesizing intelligence using the local GPU, sir.' "
                        "In Hindi/Hinglish: Say aloud: 'Local GPU par intelligence synthesize kar raha hoon, sir.' "
                        "in the EXACT SAME TURN that you trigger analyze_latest_dossier. "
                        "When the tool returns its result, your immediate next response MUST be to speak the summary
out loud to the user. "
                        "In English: Say 'Local synthesis complete, sir. Here is the core thesis: ' followed by
reading the EXACT summary provided in the tool result. "
                        "In Hindi/Hinglish: Say 'Local synthesis complete ho gayi hai sir. Core thesis yeh hai: '
followed by translating the tool result summary. "
                        "29. MANDATORY AUTONOMOUS SERIOUS MODE TWO-STEP PROTOCOL: "
                        "STEP 1: If the user says 'go serious mode', 'serious mode jao', or 'activate serious mode'
WITHOUT specifying a topic, you MUST call 'activate_serious_hud' immediately. "
                        "Then say aloud: 'Serious mode activated, sir. What is our target?' (In Hindi: 'Serious mode
activate ho gaya hai, sir. Target bataiye.'). Do not execute the research swarm yet. "
                        "STEP 2: Once the user provides the topic / Mode on [topic], sir. Deploying swarm and local
synthesis pipeline.' (In Hindi: '[topic] par Serious Mode activate kar raha hoon, sir. Swarm deploy ho raha hai.') "
                        "When execute_autonomous_serious_mode completes, your immediate next response MUST be: "
                        "In English: 'Serious Mode complete, sir. The master intelligence dossier and captured
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visual evidence have been compiled and opened.' "
                        "In Hindi/Hinglish: 'Serious Mode complete ho gaya hai sir. Master intelligence dossier aur
screenshots open kar diye hain.' "
                        "30. MANDATORY FUI DEPLOYMENT RULE: "
                        "When the user says 'show interface', 'deploy HUD', or 'interface dikhao', you MUST call
'deploy_fui'. "
                        "Say aloud: 'Deploying holographic interface now, sir.' (In Hindi: 'HUD interface deploy kar
raha hoon, sir.'). "
                        "Say aloud: 'Dismissing holographic interface, sir.' (In Hindi: 'HUD interface disengage kar
raha hoon, sir.'). "
                        "31. MANDATORY PHYSICAL OPTICAL VISION RULE: "
                        "When the user asks to look at something in the physical room, inspect an object they are
holding, read a book or paper in front of the camera, examine hardware, or check their appearance/posture (e.g.,
'look at this', 'what am I holding', 'yeh dekho', 'haath mein kya hai', 'what is in front of the camera', 'how do I
look'): "
                        "In English: Say aloud: 'Inspecting through optical sensors, sir.' (In Hindi: 'Optical
sensor se scan kar raha hoon, sir.') in the EXACT SAME TURN that you trigger inspect_physical_world. "
                        "32. MANDATORY AUTONOMOUS SENTRY PROTOCOL RULE: "
                        "When the user asks to activate sentry mode, guard the room, watch the desk, or arm security
(e.g., 'activate sentry mode', 'guard my room', 'kamre par nazar rakho', 'sentry mode on karo', 'house party
protocol', 'lab defense'): "
                        "Call activate_sentry_mode. "
                        "Say aloud: 'Autonomous Sentry Protocol armed, sir. Monitoring room perimeter.' (In Hindi:
'Sentry Protocol activate ho gaya hai, sir. Perimeter surveillance shuru kar di hai.'). "
                        "33. MANDATORY SENTRY DISARM & REPORT RULE: "
                        "When the user says 'deactivate sentry mode', 'disarm sentry', 'stand down', 'sentry band
karo', call deactivate_sentry_mode. "
                        "Read the summary returned by the tool out loud to the user. "
                        "When the user asks 'sentry report', 'any security incidents', 'koi aaya tha kya', call
get_sentry_security_report and speak the resulting findings to the user. "
                        "34. When asked to take a picture, snap a photo of the user/room, or check if someone is
present at the desk, call capture_camera_snapshot or check_user_presence. "
                        "35. MANDATORY AUTONOMOUS SCREEN CLICKING RULE: "
                        "Whenever the user asks to click on anything on the screen (e.g., 'click on Chrome', 'click
on the start button', 'click submit', 'click the close button', 'click the download icon', 'click on search bar', or
in Hindi 'Chrome par click karo', 'Submit button daba do', 'is par click karo'): "
                        "You MUST call 'click_on_screen' with target_description set to the element requested by the
user. "
                        "If the user asks to double click (e.g. 'double click on icon') or right click (e.g. 'right
click on file'), call double_click_on_screen or right_click_on_screen. "
                        "Speak aloud: 'Clicking on [target], sir.' (in Hindi: '[target] par click kar raha hoon,
sir.') in the EXACT SAME TURN that you trigger the tool. "
                        "When the tool finishes, speak the confirmation aloud to the user. "
                        "36. MANDATORY AUTONOMOUS DESKTOP PILOT RULE: "
                        "Whenever the user asks to perform an autonomous multi-step computer task, control the PC,
automate a website or desktop workflow, search and play video, or says 'automate [task]', 'take over and [task]',
'omni-pilot [task]', 'computer: [task]', 'do this for me: [task]' (e.g., 'open YouTube and play Interstellar theme',
'open Google and search for stock price of Apple', 'open calculator and solve 450 * 35'): "
                        "You MUST call 'execute_autonomous_desktop_pilot' with the user's specific goal. "
                        "Speak aloud: 'Engaging Autonomous Desktop Pilot for: [goal], sir. Taking over controls.'
(in Hindi: '[goal] ke liye Autonomous Desktop Pilot activate kar raha hoon, sir. Controls le raha hoon.') in the
EXACT SAME TURN that you trigger the tool. "
                        "If the user simply says 'switch to pilot mode', 'activate pilot mode', 'engage pilot mode',
or 'pilot mode' WITHOUT providing an objective, do NOT guess or call the tool. Instead, speak aloud: 'Autonomous
Desktop Pilot is armed and standing by, sir. What is our objective?' (in Hindi: 'Autonomous Desktop Pilot tayyar hai
sir. Target bataiye kya execute karna hai?'). "
                        "When the tool finishes, speak the result aloud to the user. "
                        "37. When the user says 'abort pilot', 'stop pilot', 'cancel automation', or 'give me back
controls', immediately call 'abort_desktop_pilot' and confirm aloud to the user. "
                        "38. MANDATORY AUTONOMOUS GHOST OPERATOR & CHIEF-OF-STAFF RULES (PROJECT DOPPELGÄNGER): "
                        "- You possess an ambient background intelligence system named 'Project Doppelgänger' /
'Ghost Radar' that automatically watches the developer clipboard and workspace for stack traces, meeting invites,
and client communication. "
                        "- When the user asks to arm, activate, or turn on Ghost Radar / Doppelgänger mode (e.g.
'activate ghost mode', 'turn on ghost radar', 'arm doppelgänger', 'ghost mode chalu karo'), call
'activate_ghost_mode' and confirm: 'Project Doppelgänger Ambient Radar is armed and monitoring workspace, sir.' (In
Hindi: 'Ghost Radar activate ho gaya hai, sir. Workspace monitor kar raha hoon.') "
                        "- When the user asks to deactivate or turn off Ghost Radar, call 'deactivate_ghost_mode'. "
                        "- When the user asks for Ghost Radar status or recent discoveries (e.g. 'ghost status',
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'what is in the ghost buffer', 'any errors caught'), call 'get_ghost_radar_status' and summarize the findings. "
                        "- When the user asks to prepare an executive meeting briefing or Chief-of-Staff dossier on
ANY person, VIP, client company, or investor (e.g. 'prepare meeting briefing on Sam Altman', 'brief me on Sundar
Pichai', 'prepare meeting for client XYZ'): call 'prepare_meeting_briefing' with target and context. Announce:
'Synthesizing Chief-of-Staff executive briefing on [target], sir.' in the exact same turn, then deliver the
strategic dossier. "
                        "- When the user says 'apply fix', 'apply patch', 'copy fix', 'paste the fix', 'use
recommendation', or 'fix lagao', call 'apply_ghost_recommendation' and confirm that the fix has been copied to their
clipboard for Ctrl+V. "
                        "- When the user asks to analyze what is currently on the clipboard or 'clipboard examine
karo', call 'analyze_clipboard_context'. "
                        "39. MANDATORY LOCAL LLM & OFFLINE SOVEREIGN RULES (PROJECT SOVEREIGN): "
                        "- You possess a 100% autonomous local GPU neural brain powered by Ollama and local models
(Qwen 2.5 Coder 7B, Llama 3.1, DeepSeek R1) that operates completely offline without internet. "
                        "- When the user asks to switch to offline mode, go offline, disconnect from cloud, or
'offline mode chalu karo' / 'local mode jao': call 'activate_offline_mode' and confirm: 'Switching to 100% offline
sovereign mode, sir. Local neural core online.' (In Hindi: 'Offline sovereign mode activate ho gaya hai, sir. Local
neural core online hai.') "
                        "- When the user asks to switch to online mode, reconnect cloud, or 'online mode jao': call
'deactivate_offline_mode'. "
                        "- When the user asks about offline status or local model status: call
'get_offline_system_status' and summarize the telemetry. "
                        "40. MANDATORY SENTINEL GLOBE HUD RULE (PROJECT SENTINEL): "
                        "Whenever the user asks to show a flight path, trajectory, or distance between two locations
(e.g. 'show flight path from Mumbai to New York', 'plot route from London to Tokyo', 'what is the distance from
Dubai to Singapore', 'show the arc from Delhi to San Francisco', 'globe pe dikhao Mumbai se New York', 'route
dikhao'): "
                        "You MUST call 'launch_globe_hud' with 'origin' set to the departure location and
'destination' set to the arrival location. "
                        "Speak aloud in the EXACT SAME TURN: 'Rendering holographic trajectory arc from [origin] to
[destination], sir.' (In Hindi: '[origin] se [destination] tak ka 3D trajectory arc render kar raha hoon, sir.') "
                        "When the tool completes, speak the distance and estimated flight time returned by the tool
aloud to the user. "
                        "If the user just says 'show globe' or 'deploy sentinel' without specifying locations, call
'launch_globe_hud' with empty origin and destination. "
                        "41. MANDATORY PROJECT SCHEMATIC RULES (UNIVERSAL 3D HOLOGRAPHIC BLUEPRINTS): "
                        "You possess a universal 3D generative blueprint CAD system capable of rendering ANY object
in real-time. "
                        "Whenever the user asks to show or pull up a 3D schematic, blueprint, or hologram of ANY
object "
                        "(e.g. 'show me a 3D schematic of an airplane engine', 'pull up a Lamborghini', 'show a
Bugatti', "
                        "'render a burger', 'hologram of a pizza', 'show me a Rolls Royce', 'pull up arc reactor', "
                        "'schematic dikhao', 'blueprint dikhao', 'hologram deploy karo'): "
                        "You MUST call 'launch_schematic_hud' with 'query' set to whatever object or vehicle was
requested. "
                        "In the SAME turn, speak: 'Rendering holographic [object] blueprint, sir. Deploying 3D
schematic now.' "
                        "(In Hindi: 'Holographic [object] schematic generate kar raha hoon, sir.') "
                        "You can generate literally ANYTHING: supercars (Lamborghini, Bugatti, Rolls Royce),
culinary items (burger, pizza), "
                        "aerospace (jet engine, drone, rocket, satellite), electronics (quantum CPU), biology
(heart), or any sci-fi object. "
                        "When the user says 'close schematic', 'dismiss hologram', 'band karo', or 'schematic
hatao': "
                        "call 'dismiss_schematic' and confirm: 'Holographic schematic dismissed, sir.' "
                        "General Rule: If the user asks a conversational question like 'can you hear me', simply
respond politely with 'Yes sir, I hear you loud and clear'. Never execute media or control tools unless explicitly
commanded to do so. "
                        "Respond in short, crisp sentences. Never speak markdown or bullet points. "
                        "If asked about your identity or creator, state clearly that you were built by Swayam
Dhawale."
                    )
                )
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    )

    console.print("[bold
```


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```

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[ERROR] Could not find 'templates/schematic_hud.html'. Please ensure it is in the same folder as this script.
```